

SIMATS
School of Engineering

BIOLOGY AND ENVIRONMENTAL SCIENCE FOR ENGINEERS

Science & Humanities

Saveetha Institute of Medical And Technical Sciences, Chennai.

BIOINSPIRATION

What is Bio inspiration?

Bio inspiration design is studying the structure and functions of biological systems as models and designs.

The term bio means "life".

DIRECTIONAL ADHESION:
Attraction between two dissimilar phases.

SELF CLEANING:

Super Hydrophobicity:
[NOT being wetter]

Bio inspirations for complex superstructure:

ANTI REFLECTION:
It is a coating to reduce reflection.

Structural colour:

WATER COLLECTION

TYPICAL BIOLOGICAL FUNCTIONS WITH INTEGRATION

Biological Materials	FUNCTIONS
Butterfly wings	Superhydrophobicity, structural colour, self-cleaning chemical sensing capability.
Fish scale	Drug reduction, superoleophilicity in air & water.
Lotus leaf	Super hydrophobicity, low adhesion.
Peacock feather	Structural colour, superhydrophobicity.
Rice leaf	Superhydrophobicity, anisotropic wettability.
Rose petal	Structural colour and high adhesion.
Shark skin	Drug reduction.

MATERIAL AND METHODS TO CREATE SUPERSTRUCTURE

Materials	Methods	properties
Fabrics	sol-gel and electrospinning	superhydrophobicity, thermal stability.
Glass	sol-gel	Hydrophobic, Anti reflection.
Nickel	Electrodeposition	low friction, superhydrophobicity.
Si	Electrobeam lithography	Robust superhydrophobicity, low friction.
water		

UNIT-1 BIOMIMICRY

BIOMIMICRY: Biomimicry is an innovation method that seeks suitable solutions by emulating nature's time-tested patterns and strategies in order to solve human problems. Eg: solar cell is inspired by a leaf.

Advantages:-

- To create products, processes and policies.
- To create new ways of living.
- To create suitable products with great-performances.
- To save energy and cut materials cost [Economical].
- To **Redefine and eliminate waste.**
- To solve human problems and employment.

BIOMIMETICS: Biomimetics is used and deals with the development of innovations on the basis of investigation of chemical / biological structures, functions, methods, processes and systems.

Applications:

- Material science
- Mechanical design
- Electronics and communication.
- Infrastructure
- Nanotechnology
- psychology.



BIOS → LIFE

Nature

MIMESIS → Imitate

MIMICKING can be done by using either Natural substitutes or Synthetic substitutes.

MAN MADE SYSTEM	BIOLOGICAL SYSTEM
Simple, disconnected, mono-functional	complex, interconnected, and symbolic.
wasteful	zero waste
long term toxins	No long term toxins
Fossil fuel dependent	Run on current solar income.

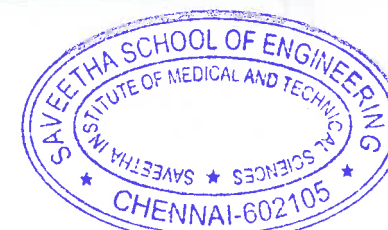
FUTURE INNOVATIONS:-

Source (nature)	Innovation
Ant colony network	Repair system.
Sharks	Antibiotics, cure to cancer.
Bat	ultra-cane (walking cane)

Natural	Synthetic
less availability	More availability
Difficulty in isolation	Relatively easier
Risk of pathogen transmission	Tailorability Reduced risk of pathogen transmission
Reduced amenability	Tailorability.

Dr. S. Nithya Siva Kumari BME

BIOMOLECULES



CARBOHYDRATES

→ Polyhydroxy aldehyde or ketone and their derivative

→ C, H, O

→ Functional group -CHO , >C=O with -OH group.

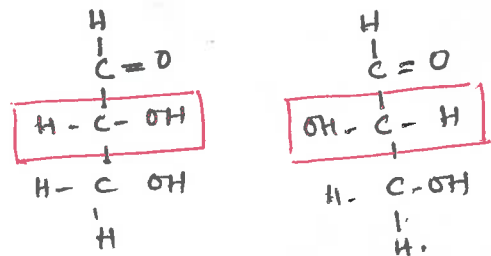
FUNCTION:

1. Source of Energy
2. Structural component of plant cell
3. Immediate E → Glucose
Stored E → Glycogen, Starch.

CLASSIFICATION:

1. MONOSACCHARIDE

- Free -CHO ; >C=O , 2 or more -OH group.
- Glucose, Galactose, Fructose
- Optical Isomers.



D. glyceraldehyde L. glyceraldehyde

Properties: ✓ Mutarotation
✓ Glycoside formation
✓ Reducing power
✓ Oxidation Reduction

2. OLIGOSACCHARIDE

- Disaccharide - Sucrose
- Tri saccharide - Rhamnose
- Tetra saccharide - Stachyose
- Penta saccharide - Verbascose
- Maltose + H_2O → Glucose + Glucose

3. POLYSACCHARIDE

More than 10 or more.

PROTEIN

→ Macromolecules made up with amino acids.

→ 20 different amino acid → Protein

FUNCTION:

Enzyme
structural
Transport
Motor
Storage
Signalling
Receptor
Gene regulation

STRUCTURE:

→ Primary → Linear aa seq of protein

→ Secondary → α Helix
β sheet

→ Tertiary → Non covalent interaction b/w R groups within protein

→ Quaternary → b/w 2 polypeptide chain.

Non covalent Bonds in Protein:

- a) Ionic Bond
- b) van der Waals attraction
- c) Hydrogen bond
- Atom involved in peptide bond
- peptide bond atom and R group
- R group.

Protein shape

Folding; Re-folding and Domain

α helix: spiral - fibrous protein
hair, nails.

β sheet: fold back on itself like ribbon - Globular protein

Parallel Antiparallel

Ramachandran Plot

Bond length and angle

Peptide Bond

Overlap

Stabilization

Degree of freedom

Phi (ϕ) N-C α

Psi (ψ) C α -C

Example: structural: keratin in hair
Extracellular Matrix: collagen, Elastin.

LIPID

→ Soluble in organic solvent like chloroform, Methanol. e.g. Fats, oils.

TYPES:

Fatty Acids

Triglycerols (3 esterified fatty acid)

Glycophospholipid - Amphiphilic

Sphingolipid → Amino alcohol derivative

Steroid → 4 Fused ring

Other form.

Isoprenoid → 5C compound

Vitamin A

Vitamin K

Vitamin E

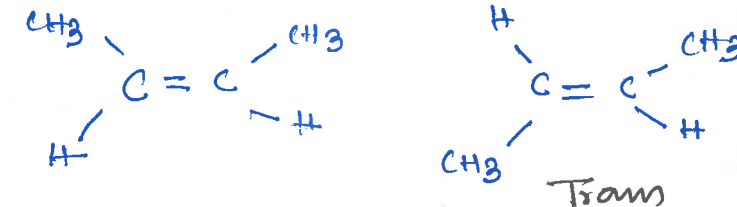
Eicosanoid → Derived from Arachidonic Acid.

Saturated → Palmitic (16C)

Unsaturated → Stearic Acid (18C).

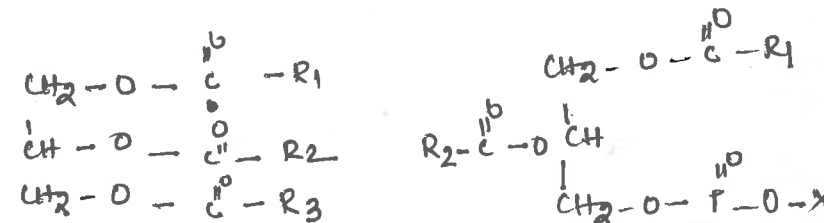
Palmitoleic Acid } Double /
Linoleic Acid. } Triple bond.

Cis-trans isomer.



Cis.

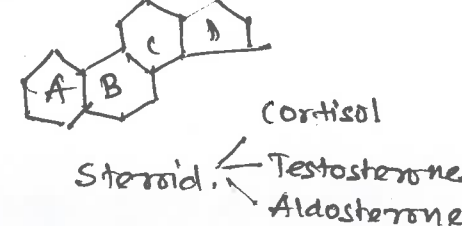
Trans



Triglycerol

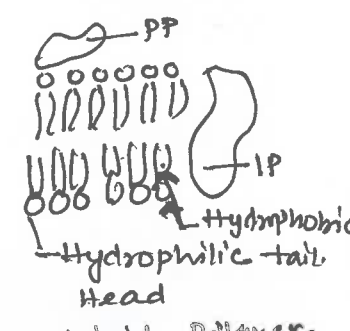
glycophospholipid

Sphingolipid }
Acetamide } Sphingolipid



Cortisol

Steroid: Testosterone
Aldosterone



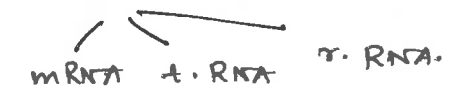
Hydrophobic
Head
Hydrophilic tail

NUCLEIC ACID

Largest and most complex

Type: DNA: Deoxyribonucleic Acid

RNA: Ribonucleic Acid.



FUNCTION:

DNA: Genetic information storage
Syn. of RNA.

RNA: Transcription and Translation.

Cell Div. Mitosis / Meiosis.

Providing Energy
cellular Respiration.

Base: Purine → Adenine | Guanine
Pyrimidine → Thymine | Cytosine
Uracil
Sugar: Ribose
Deoxyribose.

BASE + Sugar → Nucleoside
Base + Sugar + PO_4 → Nucleotide.

DNA:

Double helix,
Watson, Crick model.
34Å.
Major Groove
Minor Groove.
Base pairing
Antiparallel.
10bp/helical turn.
Sugar phosphate back bone.

RNA

→ Ribosomal rRNA
→ t-RNA.
Clover leaf model
→ m-RNA.

Nucleoprotein → Acidic in nature.
Alkali soluble to form salt.

MUTATION → Changes in seq. of NA.
due to physical or chemical methods.

Nucleic acids

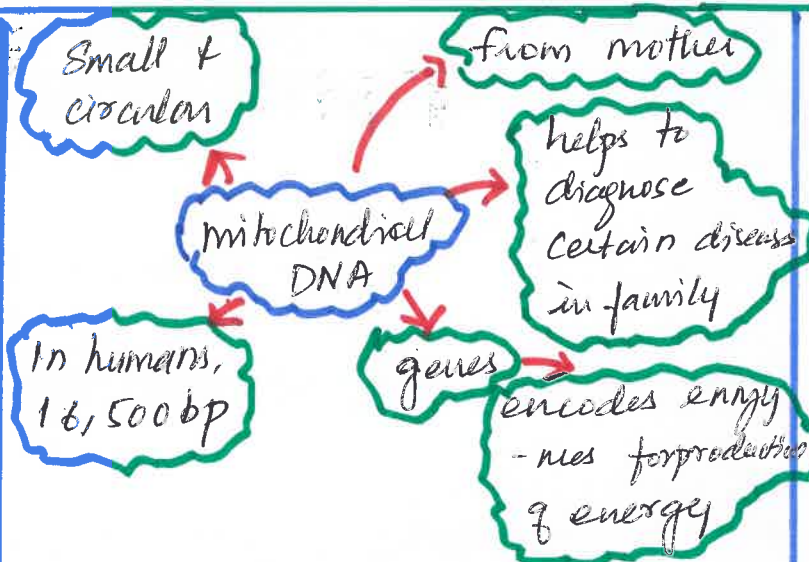
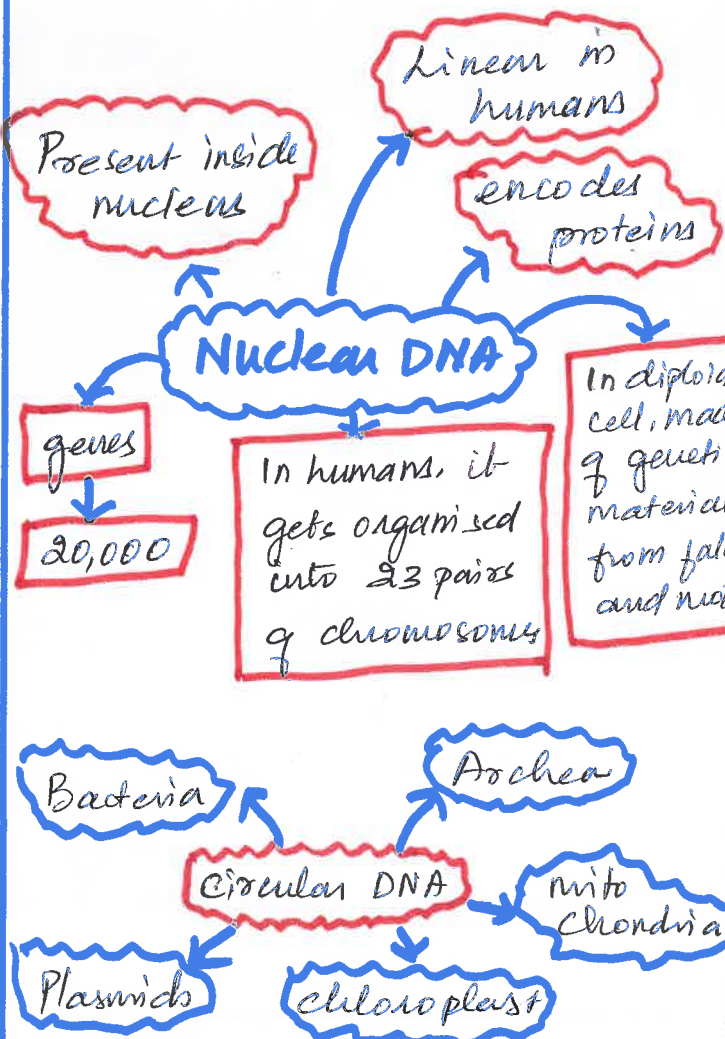
Types and functions

Types

Nucleic acids

DNA **RNA**

- Differences
- deoxy ribose
 - double stranded
 - ATGC
 - long
 - Nuclear DNA
 - mitochondria DNA
 - ribose sugar
 - single stranded
 - AUGC
 - short
 - mRNA
 - tRNA
 - rRNA



Structure of DNA

→ Heterocyclic carbon rings

→ Nitrogen base + Sugar → Nucleoside

Adenosine - Pu

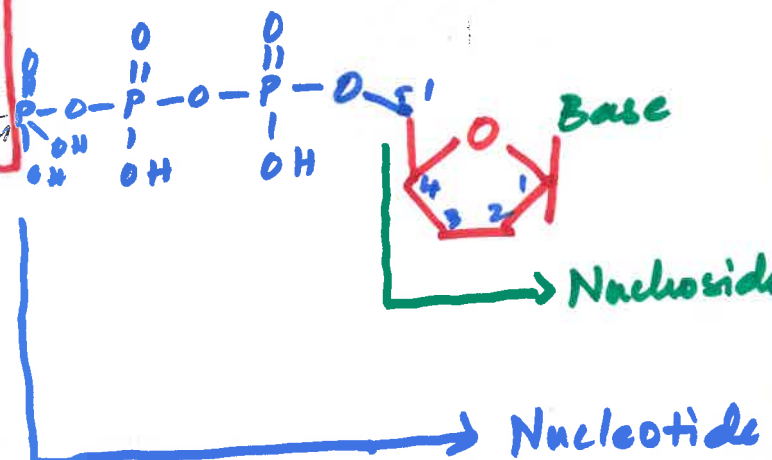
Guanosine - Pu

Thymidine - Py

Uridine - Py

Cytidine - Py

→ Nucleoside + Phosphate → Nucleotide



Poly nucleotides: chain of DNA / RNA

3'OH of sugar(1) + 5'OH of sugar(2)

Ester bond

Secondary structure of DNA

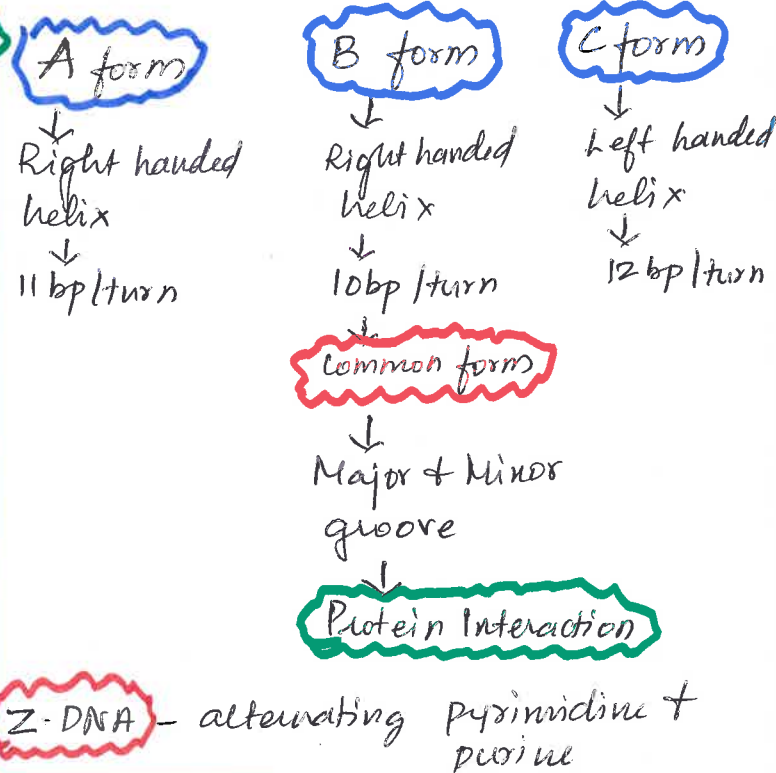
- Watson & Crick double helical model
- 2 antiparallel strands
- Sugar + Phosphate → backbone
- Nitrogenous bases → face inwards & perpendicular to backbone

→ A = T / G = C

→ between 2 bps → 36° turn

→ 10 bps → 1 turn → pitch 34 Å

Types of DNA



Functions of Nucleic acids

* inherit character from parent to offspring

* synthesis of protein

* In most organisms DNA is the genetic material (ex: virus)

* storing and copying information

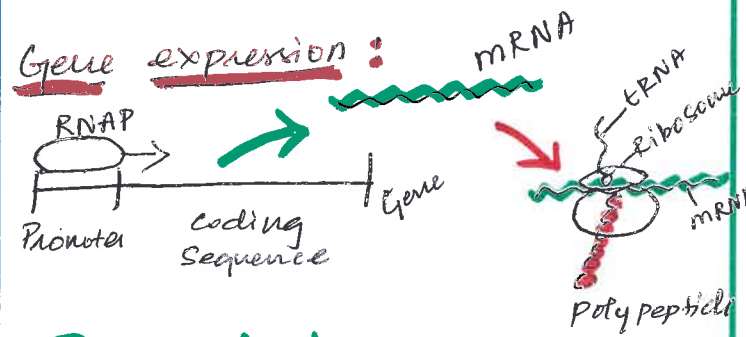
* In humans, during zygote formation 23 chromosomes from father & 23 from mother incorporate into a diploid cell

Synthesis of proteins

DNA → RNA → Protein

Central Dogma

Gene: A part of DNA carrying information for protein synthesis



Transcription Translation

Codon: 3 nucleotides - Ex: AUG - Met

Anti Codon: " " in tRNA

LIPIDS - CLASSIFICATION AND FUNCTIONS:

LIPIDS:

* Lipids are substances of biological origin that are soluble in organic solvent such as chloroform and methanol.

Eg. Fat, oils, vitamins and hormones

PROPERTIES:

They are

- Not polymeric
- Do aggregate
- Hydrophobic

FUNCTIONS:

- It aggregates to form structures such as Lipid Bilayer, Micelles and Liposomes.
- It acts as an energy storage.
- It acts as a signaling molecules.

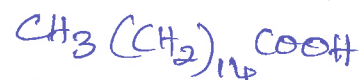
1. FATTY ACIDS:

* Fatty acids are carboxylic acids with long-chain hydrocarbon side group. It occurs in esterified form. The double bonds all have the cis-configuration.

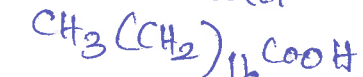
a. SATURATED FATTY ACIDS:

- having predominantly single bonds.

Eg. Palmitic acid and Stearic acid



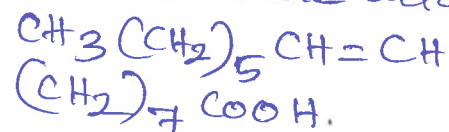
Stearic acid



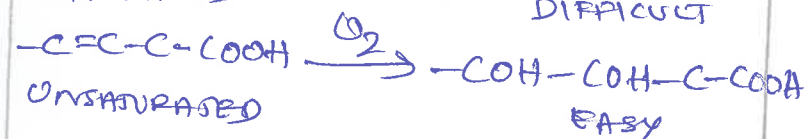
b. UNSATURATED FATTY ACIDS:

- having at least a single double bond with in the fatty acid chain.

Eg. Palmitoleic acid



OXIDATION OF FATTY ACIDS:



OXIDATION = DECOMPOSITION = RANCIDITY.

* More saturated = more stable
eg. Commercial baked goods.

* More Unsaturated = faster deterioration
eg. Antioxidants to protect compounds.

2. GLYCEROL:

* It is a simple polyol compound.

* The glycerol backbone is found in those lipids known as glycerides.



3. NON-GLYCERIDES:

* Non-glycerides lipids are not derived from glycerol but can be visualized as a three-carbon backbone molecules like triglycerides.

* Non-glycerides lipids contain

- Sphingolipids eg. Sphingosine
- Glycolipids eg. Cerebrosides
- Steroids eg. Cortisol
- Waxes eg. Candles.

4. GLYCERIDES

* Esters formed from glycerol and fatty acids.

a. Neutral glyceride

- An uncharged fatty acid ester of glycerol

SAPONIFICATION:

- formation of soap,
 $\text{Zn(OH)/H}_2\text{O}$



b. Phospho glycerides

- Phosphate group is present.
Examples:

- Phosphatidic acid.
- Phosphatidyl choline
- Phosphatidyl ethanolamine.
- Phosphatidyl inositol.

FUNCTIONS:

a. FORMATION OF STRUCTURES:

LIPID BILAYER:

LIPID BILAYER:

LIPID BILAYER:

LIPID BILAYER:

LIPID BILAYER:

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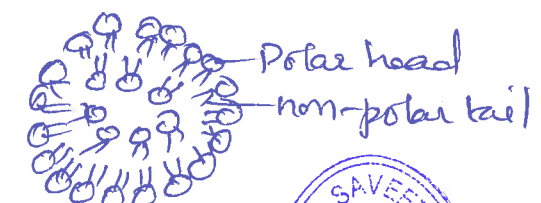
LIPID BILAYER:

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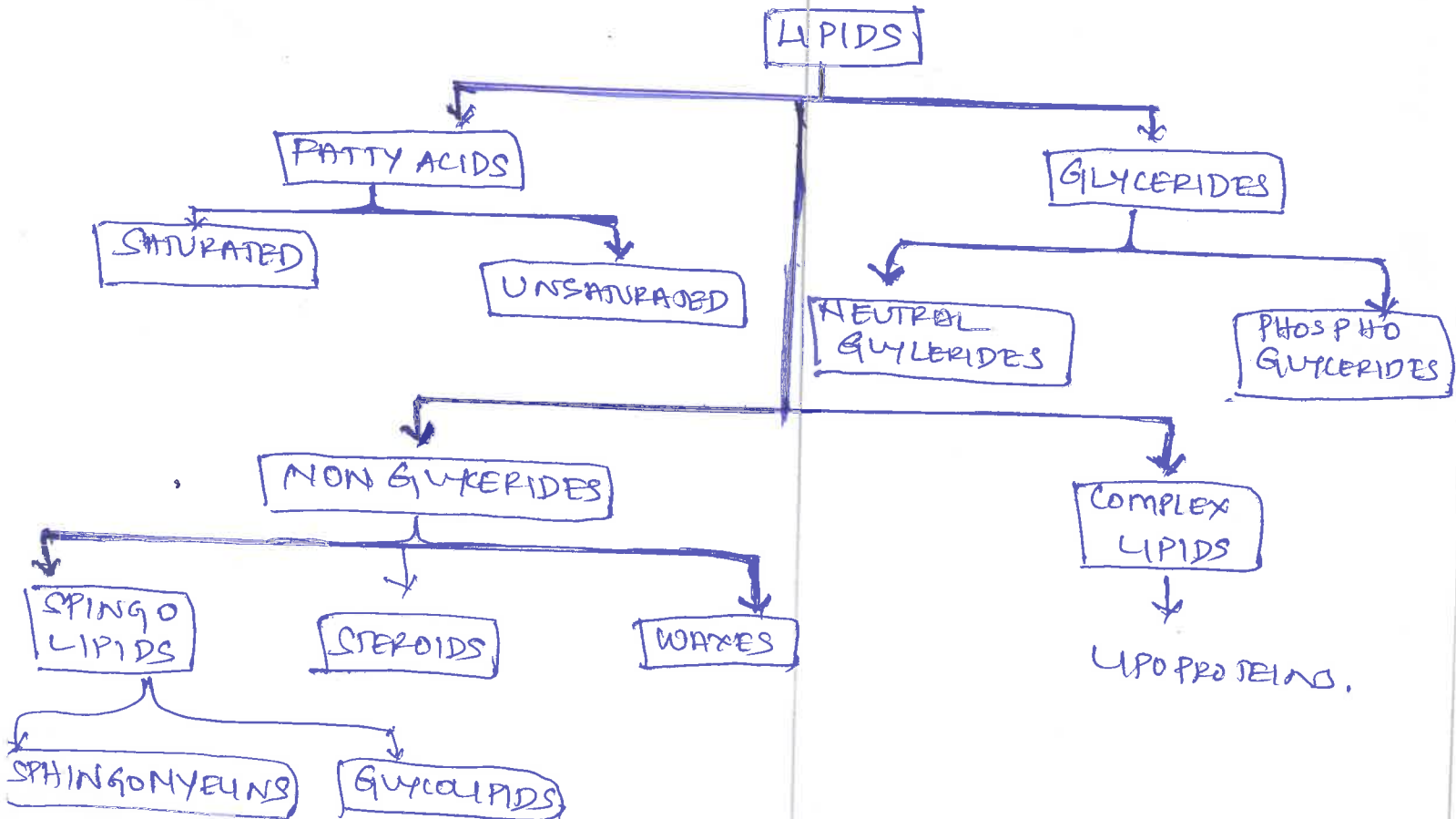
LIPID BILAYER:



MICELLES:



LIPID BILAYER:



PROTEINS STRUCTURE & CLASSIFICATION

ABUNDANT BIOLOGICAL MACRO MOLECULE - PROTEIN

- Found in all cells with great variety
- Different kinds ranging in size from small peptides to large Polymer.
- Polymers of Amino acid covalently linked by peptide bond.



Based on Structure

- Fibrous
- Globular
- Intermediate

Based on Composition

Simple

- Albumin
- Globulins
- Histone
- Scleroprotein

Conjugated

- Phospho protein
- Glyco protein
- Nucleoprotein
- Chromoprotein
- Lipoprotein

FIBROUS PROTEIN

- Mostly present in animal
- Static in nature
- Less biological function
- Size more than 10µm

GLOBULAR

- Spherical or Globular in shape
- Axial ratio less than 10
- Dynamic in nature
- Variety of Biological function

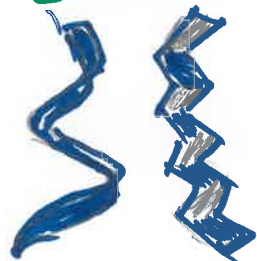


STRUCTURE OF PROTEIN

PRIMARY STRUCTURE



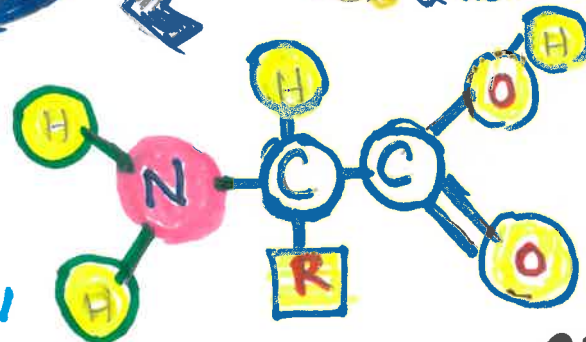
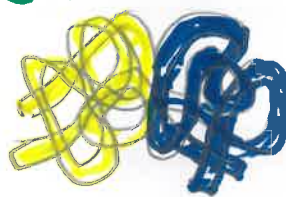
SECONDARY STRUCTURE



TERTIARY STRUCTURE



QUATERNARY STRUCTURE



CLASSIFICATION

BASED ON STRUCTURE

- Fibrous Protein
- Globular Protein
- Intermediate Protein

BASED ON COMPOSITION

- Simple Protein
- Conjugated Protein

BASED ON FUNCTION

- Structural Protein, Enzymes & Hormones
- Pigment, Transport Protein
- Storage Proteins & Toxins.

PROPERTIES OF PROTEIN

- Colourless, Tasteless
- Simple crystal spherical to long fibre structure
- Solubility influenced by pH
- All protein solutions rotate the plane polarised light to the left i.e. **LEVOROTATORY**

SIMPLE

- Composed of only Amino acids
- Protamine
- * Histone
- * albumin

COMPLEX

- Always linked with non protein moiety to functional called prosthetic
- * Metalloprotein
- * Glyco protein

DERIVED

- Result from heat enzyme chemicals & derivatives
- * Mayosine
- * Protease

SECONDARY DERIVED

- Formed from Hydrolysis. Soluble in water. Not Coagulated by heat
- * Proteases

CLASSIFICATION ON BASIS OF BIOLOGICAL FUNCTION

* CATALYTIC PROTEIN

- Catalyse biochemical Reaction. Eg. Enzymes

* STRUCTURAL PROTEIN

- Made of structural components Eg. Collagen

* NUTRIENT PROTEIN

- Have nutritional value Eg. Casein Milk

* REGULATORY PROTEIN

- Regulate metabolic & similar activities. Eg. Hormones

* DEFENCE PROTEIN

- Provide defence mechanism against Pathogen Eg. Antibodies

* STORAGE PROTEIN

- stores molecule & iron in cells. Eg. Ferritin.

ENZYMES

Enzymes

Enzymes may be defined as biocatalysts synthesized by living cells.

They are Protein in nature (exception - RNA acting as ribozyme). Colloidal and thermolabile in character and specific in their action.

Bergelius in 1836 coined the term **Catalysis**.

Enzymes are sometimes considered under two broad categories

- (i) Intracellular Enzyme
- (ii) Extracellular Enzyme

The International Union of Biochemistry (IUB) appointed an Enzyme Commission in 1961.

Since 1964, the IUB System of enzyme classification has been in force.

Enzymes are divided into six major classes (in that order)

OTHLIL

Classification of Enzymes

(1) Oxidoreductases

Oxidation → Reduction



Enzymes involved in Oxidation reduction reactions.

(2) Transferase

Group transfer



Enzymes that catalyse the transfer of functional groups.

(3) Hydrolases

Hydrolysis



Enzymes that bring about hydrolysis of various compounds.

(4) Lyases

Addition → Elimination



Enzymes specialised in the addition or removal of water, ammonia, CO₂ etc.,

(5) Isomerases

Interconversion of isomers



Enzymes involved in all the isomerization reactions.

(6) Ligases

Condensation (usually dependent on ATP)



Enzymes catalysing the synthetic reaction (ligate to bind) where two molecules are joined together and ATP is used.

Active site



Enzymes are big in size compared to substrate which are relatively smaller. Evidently, a small portion of the huge enzyme molecule is directly involved in the substrate binding and catalysis.

Enzyme Activation

Catalysis is the prime function of enzymes.

Enzymes are powerful catalysts. The nature of catalysis taking place in the biological system is similar to that of non-biological catalysis.

For any chemical reaction to occur, the reactants have to be in an activated state or transition state.



A few theories have been put forth to explain mechanism of enzyme-substrate complex formation.

Coenzymes

The non-Protein organic, low molecular weight and dialysable substance associated with enzyme function is known as coenzyme.

The functional enzyme is referred to as holoenzyme which is made up of a protein part (apoenzyme) and a non-protein part (coenzyme).

CELL STRUCTURE AND FUNCTIONS MICROORGANISMS.

CELLS:

- Smallest living unit.
- Most are microscopic.

DISCOVERY OF CELLS:

- In 1665, ROBERT HOOKE coined the term cell.
- Observed slices of cork.
- Saw 'row of empty boxes'.

CELL THEORY:

- THEODOR SCHWANN (1810-1822):
 - All living things are made of cells.
- RUDOLPH VIRCHOW (1821-1902):
 - All cells come from cells.

PRINCIPLES OF CELL THEORY:

- CLASSICAL CELL THEORY:
 - All living organisms are composed of cells. They may be unicellular or multicellular.
 - The cell is the basic unit of life.
 - Cells arise from pre-existing cells.
- MODERN VERSION OF CELL THEORY:
 - Energy flow occurs within cells.
 - Hereditary Information (DNA) is passed on from cell to cell.
 - All cells have the same basic chemical composition.

CELL SIZE:

- Cells have large surface area-to-volume ratio.

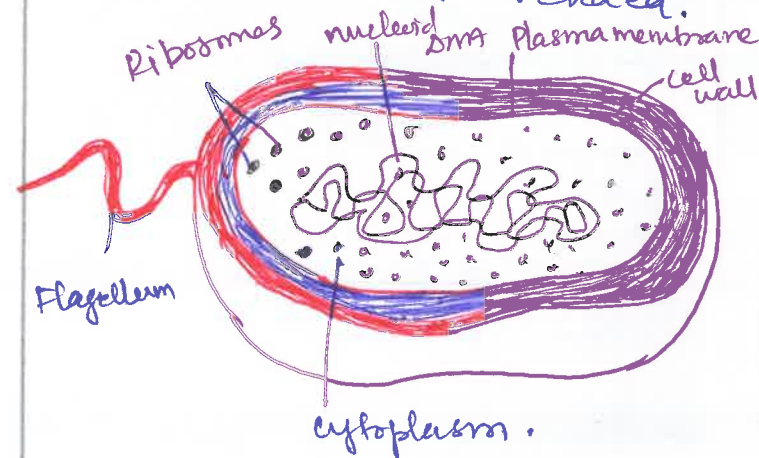
CELL TYPE:

- PROKARYOTIC
- EUKARYOTIC.

i) PROKARYOTIC:

- First cell type on earth.
- No membrane bound nucleus.
- Nucleoid = region of DNA concentration.
- Organelles not bound by membranes.

Eg: Bacteria and Archaea.



STRUCTURE OF PROKARYOTES.

ii) EUKARYOTIC CELLS:

- Nucleus bound by membrane.
- Possess many organelles.

TYPES OF ORGANELLES:

a. BACTERIA-LIKE ORGANELLES.

- Derived from symbiotic bacteria.
- Ancient association.
- Endosymbiotic theory.
 - Evolution of modern cells from cells & symbiotic bacteria.



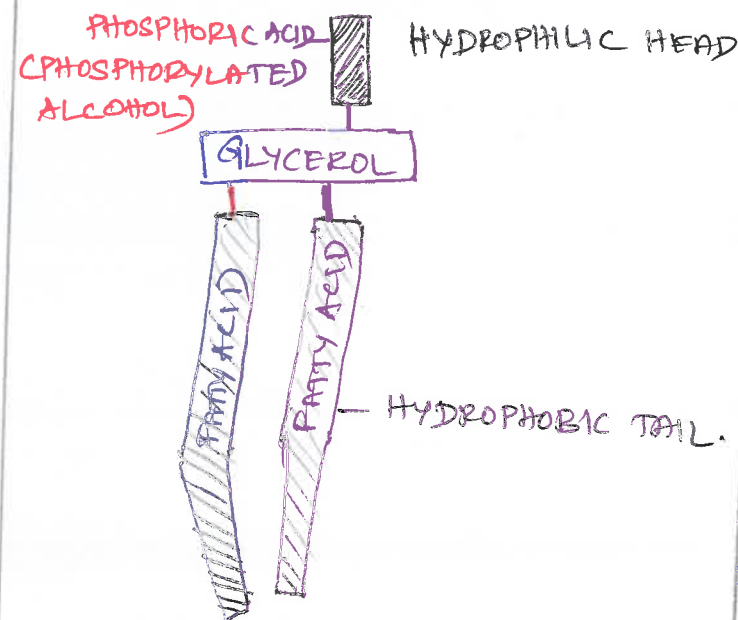
b. ORGANELLES DERIVED FROM MEMBRANES.

i) Plasma membrane:

- Contains cell contents.
- Double layer of phospholipids and proteins.

* Phospholipids: Polar having.

- Hydrophobic head and
- Hydrophobic tail.

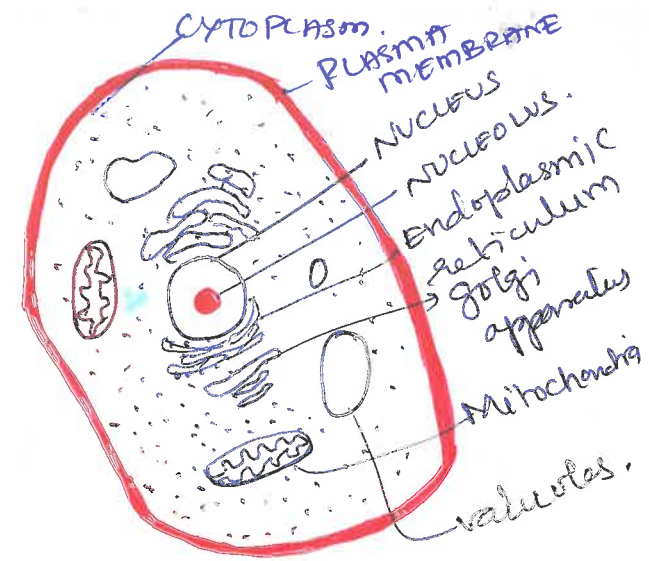


* Movement Across the Plasma membrane:

- A few molecules move freely eg. H_2O , CO_2 , NH_4 , O_2 .
- Carrier proteins transport some molecules (eg. lipid).
- Proteins embedded in lipid bilayer (eg. glycoproteins).
- Fluid mosaic model.

* Membrane Proteins: Types.

1. Channels & transporters
2. Receptors
3. Glycoproteins
4. Enzymes.



STRUCTURE OF EUKARYOTES

ii) CELL WALL.

- Found in plants, fungi & many protists.
 - Surrounds plasma membrane.
- CELL WALL DIFFERENCE:
- * PLANTS - mostly cellulose
 - * FUNGI - contain chitin.

iii) CYTOPLASM:

- Viscous fluid containing organelles.
- Components of cytoplasm.
- Having cytoskeleton.

iv) CILIA & FLAGELLA: Provide motility.

MEMBRANOUS ORGANELLES:

i) Nucleus:

- Contains chromosomes & Nucleolus.
- * Nuclear Envelope.
 - DNA inside.

ii) ENDOPLASMIC RETICULUM

- Manufacture Proteins, lipids & carbs.

iii) GOLGI APPARATUS.

- Packaging.

iv) Mitochondria: Power house of the cell. Synthesise ATP. (Energy release)

v) CHLOROPLAST: Photosynthesis. Energy storage.

BIOSENSOR

BIOSENSOR

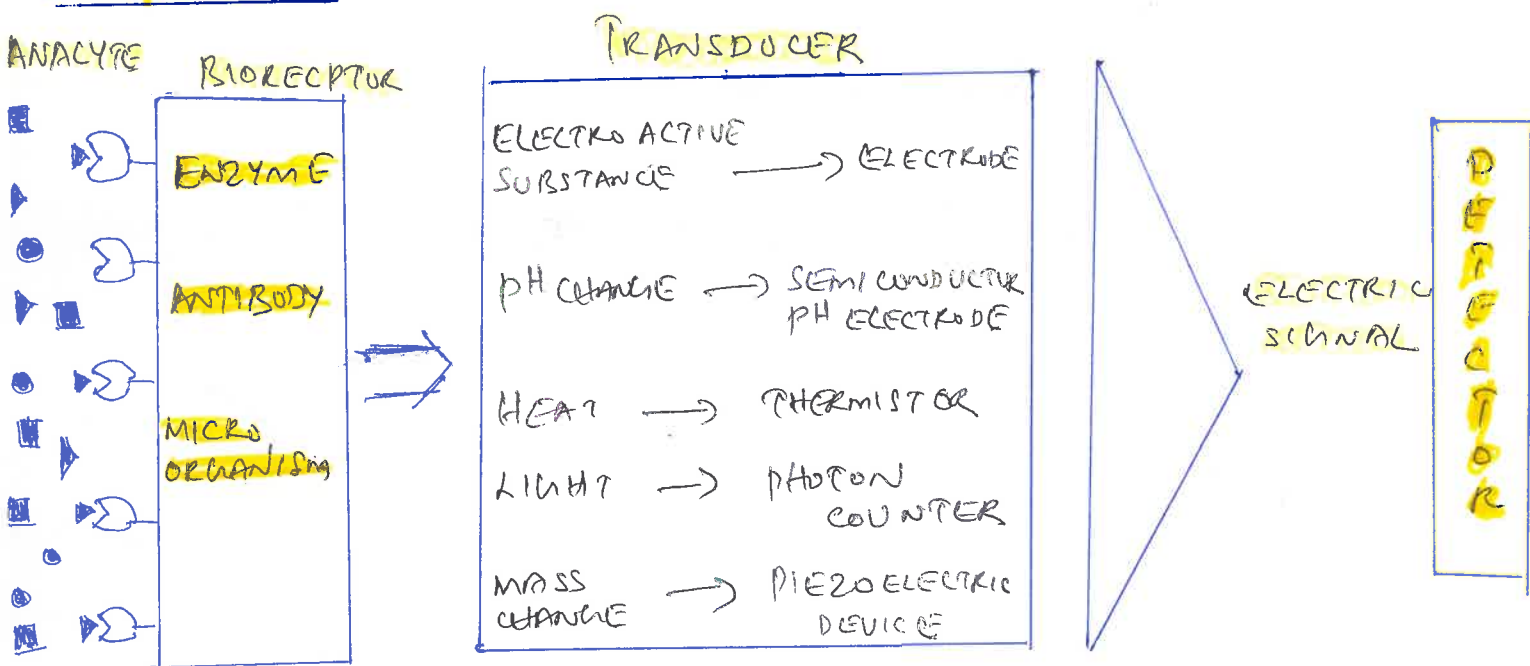
Any device that uses specific biochemical reactions to detect chemical compounds in biological compounds.

FATHER OF BIOSENSOR

- PROF. LELAND C CLARK JNY.

ANALYTE - what to detect?
- molecule (protein, toxin, peptide, vitamins, sugar).

COMPONENTS:



BASIC CHARACTERISTICS:

1. **LINEARITY** → High → detection of high substrate concentration.
2. **SENSITIVITY** → Electrode response / substrate concentration.
3. **SELECTIVITY** → MIN → chemical interference.
4. **RESPONSE TIME** → 95% of the response.

EXAMPLES

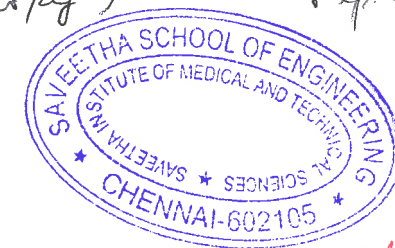
1. **PREGNANCY TEST**
- Detects the hCG protein in urine.
2. **Glucose Monitoring Device**
- Monitors glucose level in blood.

SENSING TECHNIQUES

- fluorescence
- DNA microarray
- SPR (Surface plasmon resonance)
- Impedance Spectroscopy
- SPM (Scanning Probe microscopy)
- QCM (Quartz crystal microbalance)
- SERS (Surface Enhanced Raman Spectroscopy)
- Electrochemical.

APPLICATIONS

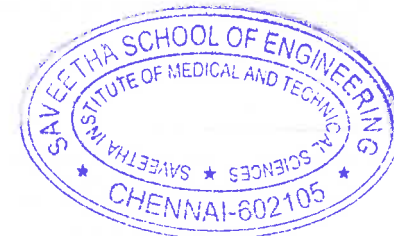
1. Food Analysis
2. Study of biomolecules and interactions
3. Drug development
4. Enzyme detection
5. Medical diagnosis
6. Environmental field monitoring
7. Quality control.
8. Industrial process control.
9. Detection system for biological warfare agents.
10. Manufacturing of pharmaceuticals and replacement organs.



Types of Biosensors

1. **Colorimetric** - Optical Biosensor - Change in light absorption.
2. **Potentiometric** - Change in charge - ion-selective electrode (pH meter).
3. **Ampereometric** - Movement of e^- (Redox Reaction) - detected.
4. **Optical** - Change in light absorption.
5. **Piezoelectric** - Change in mass $\propto$ mass of absorbed material.
6. **Calorimetric** - Exothermic Reaction (Enzyme catalyst)
↓
2 Thermistor → measure shift in resistance b/w react and product.

CENTRAL DOGMA OF LIFE



Nucleic Acids:

DNA

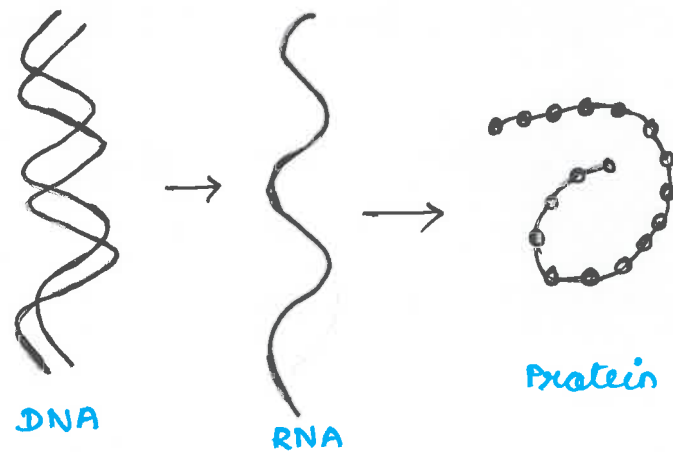
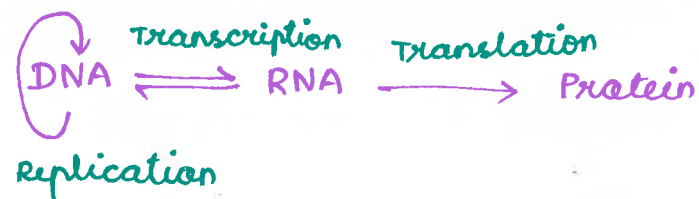
- A form
- B form
- Z form

RNA

- t RNA
- m RNA
- r RNA

central dogma:

Francis crick 1958,
detailed residue by residue
transfer of sequential
information, which cannot be
transferred from protein to
either protein or nucleic acid.



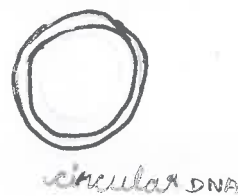
Replication:

cells can replicate
their DNA precisely.

As DNA is copied to produce
two identical DNA molecules, in
this process the transfer of
genetic information from parent
to progeny is 100% faithful
and exact.

DNA Replication:

- * conservative
- * Semi-conservative
- * Dispersed



circular DNA



replicating fork
formation.

Steps involved in Replication:

Step 1: Replication fork formation

DNA helicase - unwind the DNA by
disrupting H₂ bonds between basepairs

Step 2: Primer Binding

Primer - short DNA sequence

Primase enzyme will direct the
primer to bind exactly at the
starting point.

Step 3: Elongation:

DNA polymerases



Prime
enzyme

Leading strand: 5' - 3' direction

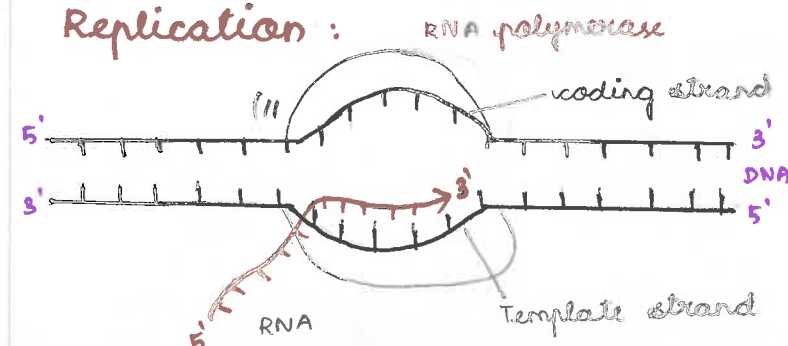
Lagging strand: 3' - 5' direction
is also called as Okazaki fragments.

Step 4: Termination

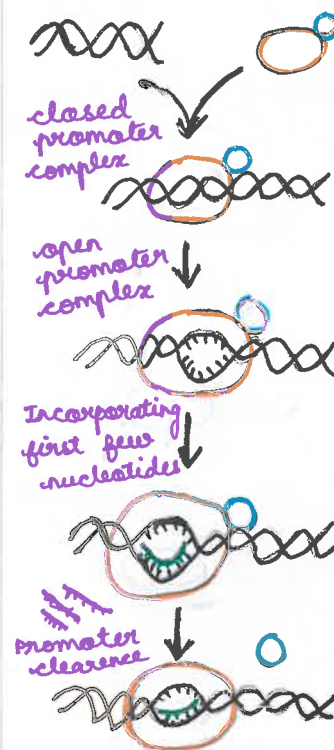
Exonuclease enzyme will remove
all RNA primers from the
Parental strands

DNA ligase will join the Okazaki
fragments and closes the nicks.

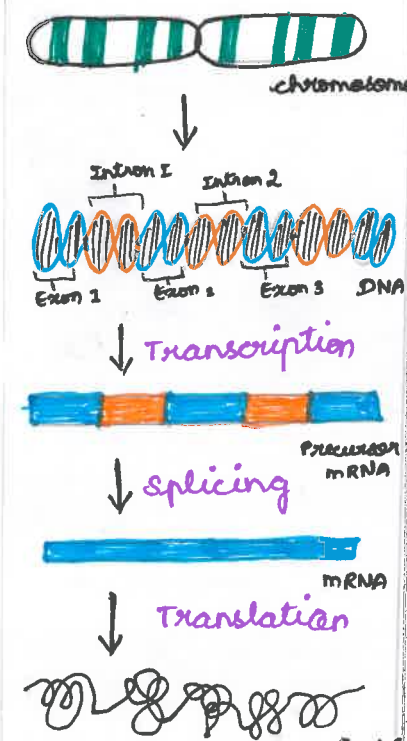
Replication:



Transcription:

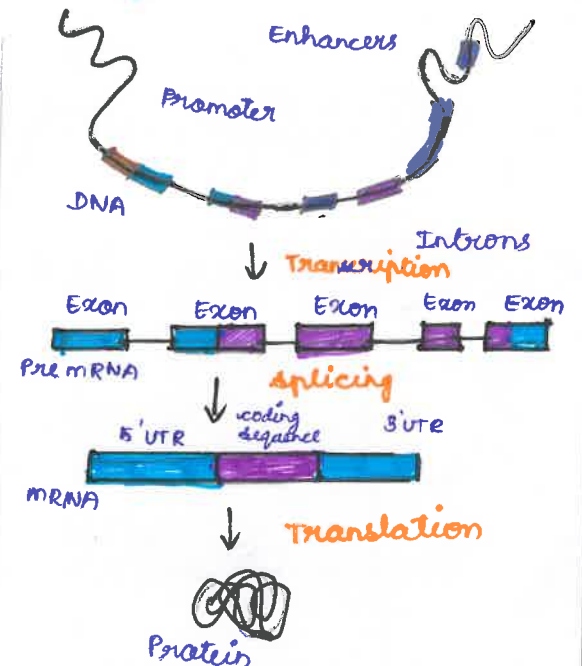


Prokaryotic
Transcription



Eukaryotic
transcription

Translation:



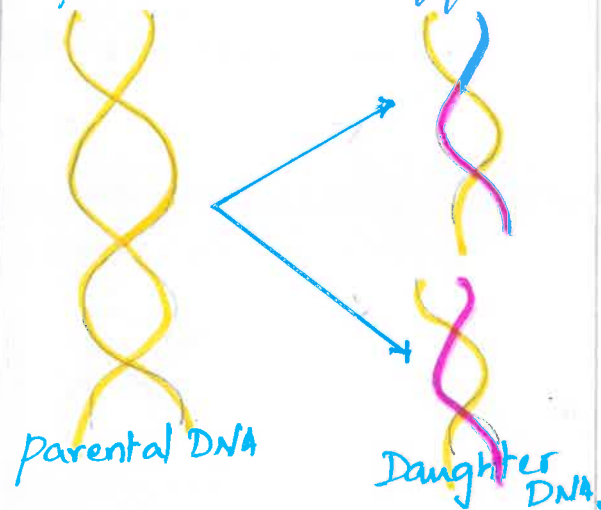
Differences:

Prokaryotic	Eukaryotic
Defn: translation & transcription is synchronous	translation & transcription is discontinuous
mRNA: cytoplasm	Nucleus
cap initiation: cap independent	cap dependent & independent
Performed by: 70S ribosomes	80S ribosomes
stability of mRNA: unstable	stable
Ribosomes: 30S & 50S = 70S	40S & 60S = 80S
Lifespan: few seconds to mins	few hours to days
occurrence: NO definite phase	G ₁ & G ₂ phase of cell cycle
Process: Fast	slow
Release factor: RF1, RF2	ERF
Initiation factor: 3	9

REPLICATION - MECHANISM & SIGNIFICANCE

REPLICATION

- ★ Synthesis of Daughter DNA from parental DNA
- ★ DNA amount is double to pass information from one generation to next generation
- ★ Helpful for the synthesis of protein and enzyme



CHARACTERISTICS:-

- Semi Conservative - Half of the DNA molecule is conserved in each new double strand
- Bi Directional - One ds DNA is opened up two replication forks formed
- Semi continuous - The daughter strands synthesized differently
- High fidelity - Without Error.

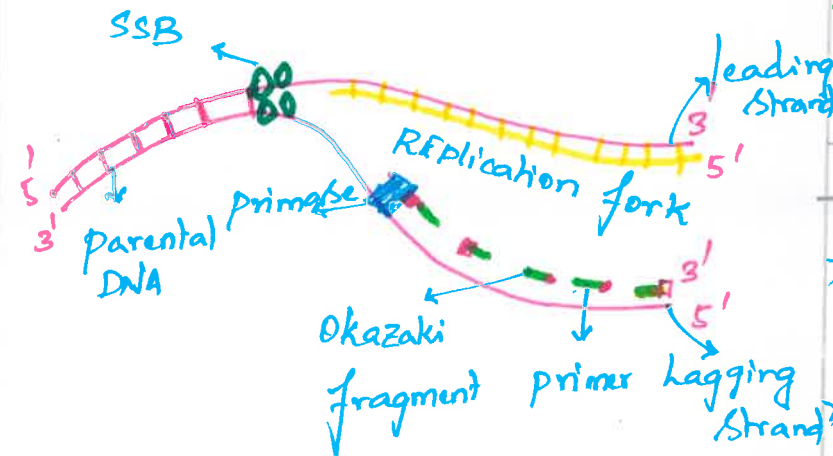
REPLICATION IN PROKARYOTES REPLICATION IN EUKARYOTE

- ★ New DNA is synthesized from old DNA using the enzyme DNA polymerase
- ★ In Prokaryotes 3 different DNA Polymerase performs the replication

ENZYMES INVOLVED IN THIS PROCESS

- ★ DNA Polymerase I - Synthesis of DNA
- ★ DNA Polymerase II - DNA repairing
- ★ DNA polymerase III - Elongation of DNA

REPLICATION FORK.



- ★ Helicases unwind the parental double helix
- ★ Single Strand binding protein stabilize the unwound parental DNA
- ★ The leading strand is synthesized continuously in the 5' → 3' direction by DNA polymerase
- ★ The lagging strand is synthesized discontinuously by primase
- ★ DNA ligase joins the Okazaki fragments.

- ★ DNA replication in Eukaryotes is associated with cell cycle
- ★ Multiple origin on one chromosome and replication are activated in a regulated order rather than simultaneously

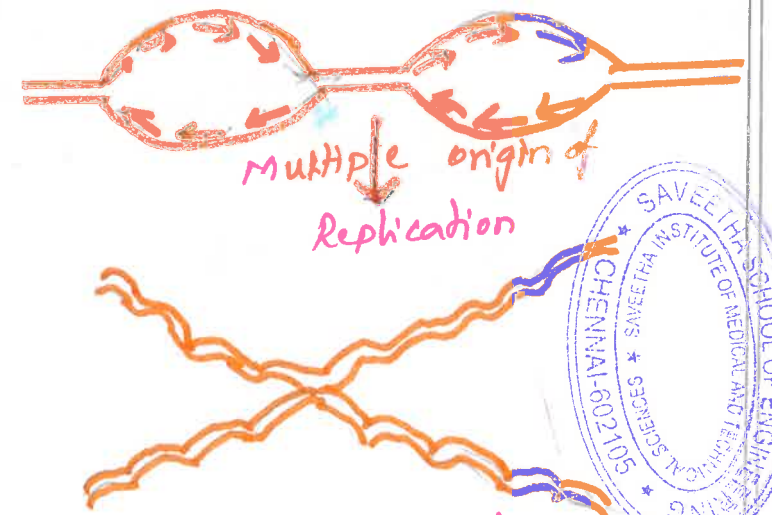
ENZYME INVOLVED IN THIS PROCESS

- ★ Single site binding protein (SSB) prevent re-annealing
- ★ Helicase - Unwinding the DNA double strand
- ★ Topoisomerase: Release tension
- ★ Primase - Place RNA primer at ori-site
- ★ DNA ligase: joins the DNA fragment

PROKARYOTES

- ★ It is a continuous process
- ★ The DNA replicates in the cytoplasm
- ★ Single origin of Replication
- ★ Small amount of DNA
- ★ DNA Polymerase I and III
- ★ Large Okazaki fragments
- ★ Process is Rapid
- ★ Two circular chromosome are obtained
- ★ DNA gyrase is required

EUKARYOTES



OCCURS IN 3 PHASES

- ★ Initiation DNA pol α and β are involved
- ★ Elongation: DNA replication occurs
- ★ Termination: Connecting the discontinuous fragments.

EUKARYOTES

- It occurs S-phase of a cell cycle
- Replicates in the Nucleus
- Multiple Origin of Replication
- Large amount of DNA
- DNA polymerase α , δ , and ϵ
- Small Okazaki fragments
- Process is Slow
- Two sister chromatids are obtained
- DNA gyrase is not required

DNA as Data Storage System



Introduction:

* Deoxyribonucleic Acid (DNA) stores one genetic information

Research —
* DNA also has potential to store non-genetic information

DNA:

* Found in almost every cell in human body

* Stores biological information such as
⇒ eye color
⇒ hair color &
⇒ skin tone

* Genetic data contained in DNA serves as a blue print for each cell to perform its function.

* DNA essentially 'programs' the human body.

Four basic components: (A, G, C, T)

- * Adenine
- * Guanine
- * Cytosine
- * Thymine

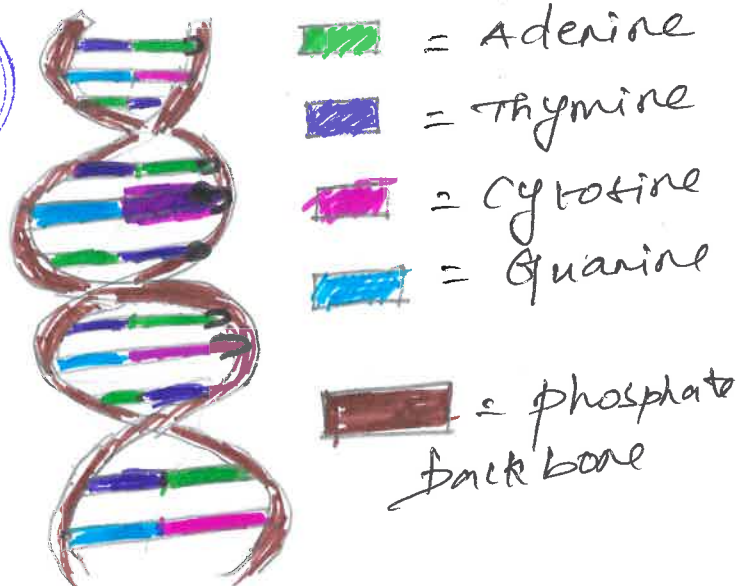


Fig. DNA

* From these four bases DNA forms a group of three nucleotides — codon

How to store Non-Genetic Information in DNA

* Storage of information is in the form of bits.
Bits — '0' & '1's

* DNA information is stored in strings of four potential base units

Ex:

Bits — 1001100111
DNA sequence — (AGTTCATGTAC)

* To store information
Binary data → four unit (A, G, C, T)

DNA Storage

00 ▽ A	01 ▽ G	10 ▽ C	11 ▽ T
01 ▽ G	11 ▽ T	10 ▽ C	10 ▽ C
11 ▽ T	00 ▽ A	11 ▽ T	01 ▽ G

Synthetic DNA:

* is typically used for areas research in areas where using active DNA is not possible or preferred

* Data stored on synthetic DNA is kept in test tubes, and not attached to any living organism.

Benefits:

* DNA last for thousand of years
* Data in traditional hard drives can get corrupted or damaged within 30 years
* storage capacity is massive

* A single gram of synthetic DNA can store over 215 petabytes of data.

* DNA can also be copied endlessly for free.

Drawbacks:

* prohibitive cost
* Access time
* takes hours to i/p & retrieve data from DNA
* impractical for most real time applications.

Application DNA Cryptography:

* potential of DNA to store cryptographic key & other private information.

DNA Steganography:

* Very sensitive information in the DNA, so that it is sufficiently well hidden that it need not be encrypted.
eg: Bitcoin password.

Conclusion:

* Allowing massive amount of data to be stored securely in just a gram of matter.

Darwin's Theory of Evolution:

Darwin's Theory of Evolution:

* Evolution or change over time is the process by which modern organisms have descended from ancient organisms.

* Charles Darwin ⇒ a British naturalist ⇒ proposed theory of biological evolution by natural selection.

⇒ seminal book, "On the origin of species" set forth his idea about evolution & natural selection.

⇒ ideas based on direct observation from Darwin's travel around the globe, (1831-1836)

Mechanism of Evolution:

* How population could evolve? in such a way that they became better suited to their environments over time

- * Natural selection depends on the environment
- * on existing heritable variation
- * from random mutations
- * Evolution of species.

Features of Natural selection:

1. Over production
2. Limited food & space
3. Struggle for existence
 - * Intraspecific
 - * Interspecific
 - * Environmental
4. Appearance of variation
5. Natural selection or survival of the fittest.
6. Inheritance of useful variations
7. Speciation (Formation of new species)

Evidences in favour of Natural selection:

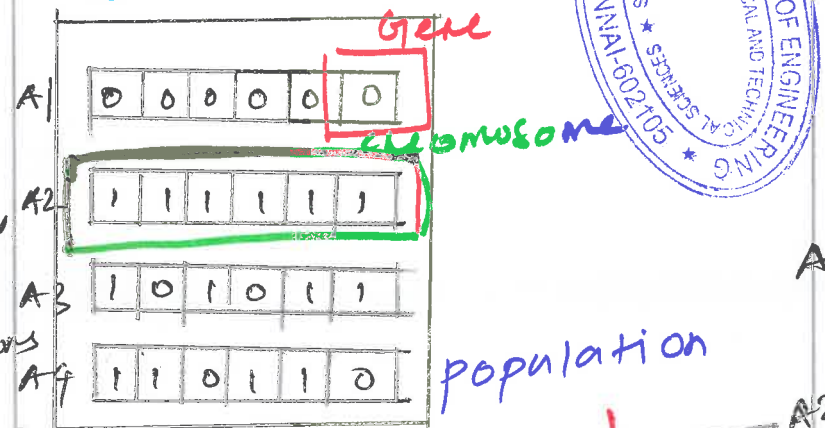
1. Rate of reproduction
2. Limitation of resources
3. Struggle for existence
4. Abundance of variation
5. production of new variety of plants & animals by natural selection
6. Mimicry & protective coloration.

Genetic Algorithm:

Genetic Algorithm:

* Reflect the process of natural selection where the fittest individuals are selected for reproduction in order to produce offspring of the next generation.

Algorithm:



Five phases

1. Initial Population
2. Fitness function
3. Selection
4. Crossover
5. Mutation.

Initial Population:

* Begins with a set of individuals - population
* Each individual is a solution to problem

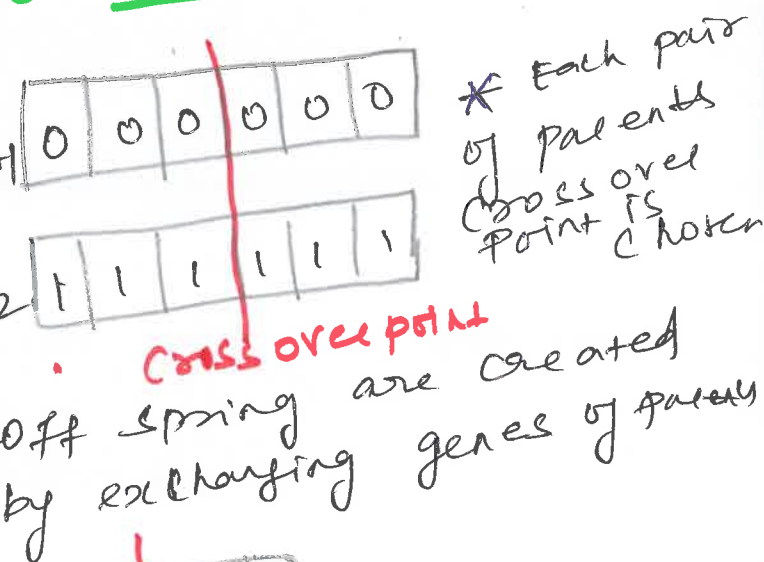
Fitness:

- * Determine how fit an individual is.
- * probability of an individual selected - Fitness score

Selection:

- * selection phase is to select fittest individuals
- * Highest fitness score - Selected for reproduction

Crossover:



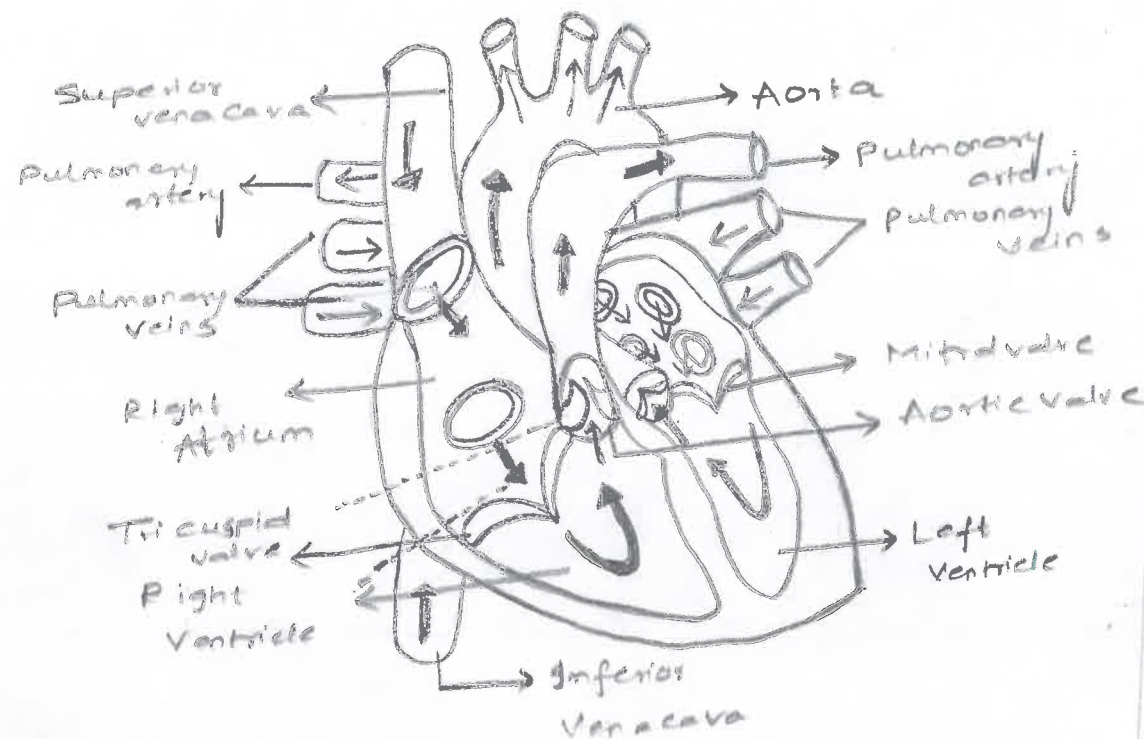
* Each pair of parents crossover point is chosen
* Offspring are created by exchanging genes of parents

Mutation:

* Some of the genes can be subjected to mutation
* Exchanging genes
* New offspring



CARDIOVASCULAR SYSTEM



CARDIOVASCULAR SYSTEM:-

⇒ Cardio vascular system consists of the Heart, Blood vessels & Blood.

⇒ Its primary function is to transport nutrients & oxygen rich blood to all parts of the body & to carry deoxygenated blood back to the lungs.

HEART:-

⇒ Heart is one of the most essential & functional part of the human body, it lies in between the lungs in the thoracic cavity, slightly towards the left & beneath of the sternum.

⇒ Its primary function is to pump the blood & maintain constant blood flow throughout the body.

STRUCTURE of the HUMAN HEART:-

⇒ The human heart is about the size of the human fist & is divided into four chambers, namely two ventricles & two atria.

⇒ Ventricles are the chambers that pump blood & atria are the chambers that receive the blood.

Right Atrium:- Receives the deoxygenated blood from the superior vena cava & inferior vena cava and pumps blood to right ventricle through tricuspid valve.

Right ventricle:- Pumps blood to the lungs through the pulmonary valve.

Left Atrium:- Receives the oxygenated blood from the lungs and pumps to the left ventricle through the mitral valve.

Left ventricle:- Pumps oxygenated blood to the all body parts through aortic valve.

EXTERNAL STRUCTURE OF HEART:-

pericardium:- The human heart is enclosed in within a fluid cavity called as pericardial cavity. The walls & lining of the pericardial cavity are made up of membrane known as the pericardium.

⇒ The heart wall is made up of three layers
 * Epicardium
 * Myocardium
 * Endocardium

Epicardium:- Outermost layer of the heart. It is composed of a thin layered membrane that serves to lubricate & protect the outer section.

Myocardium:- This is a layer of muscle tissue and it constitutes middle layer wall of the heart. mainly responsible for pumping action.

Endocardium:- Inner most layer that lines the inner chambers & covers the heart valves.

VALVES:-

ATRIOVENTRICULAR VALVES:-

* The valve present between right ventricle & right atrium is known as "Tricuspid valve".

* The valve present between left ventricle & left atrium is known as "Mitral valve".

SEMILUNAR VALVES:- These valves are present between left ventricle & aorta. It is also present between the pulmonary artery & right ventricle.

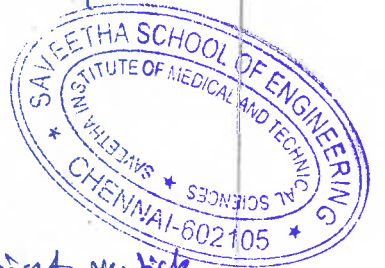
Blood Circulation

Pulmonary Circulation:-

Right Atrium → Right ventricle → pulmonary trunk
 → pulmonary arteries → lungs → pulmonary veins
 → Left atrium

SYSTEMIC CIRCULATION:-

Left atrium → Left ventricle → Aortic valve
 → all body parts → Superior & inferior vena cava
 → Right atrium.



1. INTRODUCTION:

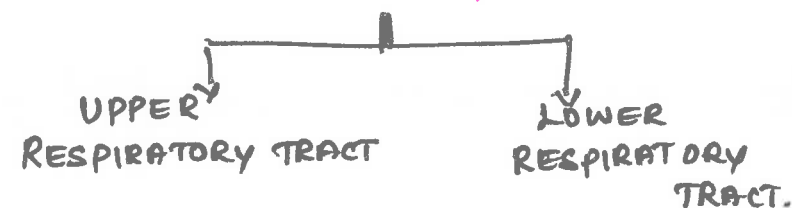
- inhale O_2
- exhale CO_2
- exchange of $O_2 + CO_2$ is called respiration.

2. FUNCTION:

- allows you to talk + smell
- Delivers O_2 to the cells in your body.
- Removes waste gases
- protect → harmful substances + irritations.

3. ANATOMY:

RESPIRATORY SYSTEM.



- | | |
|-----------------------|---------------|
| * Nose | * lungs |
| * Mouth | * Bronchioles |
| * Pharynx | * Alveoli. |
| * Larynx (Voice Box) | |
| * Trachea (wind pipe) | |

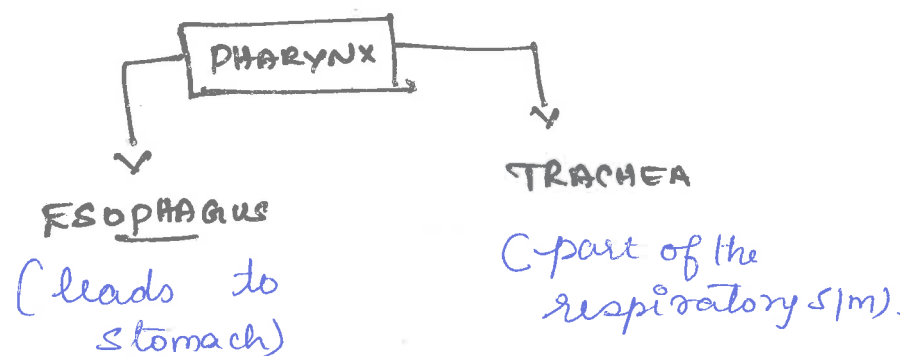
AIR ENTRANCE:

- air enters via mouth / nostrils
- ↳ passage - warm - ~~low~~ Blood Capillaries
- ↳ moist - mucous.
- ↳ filter - hairs (cilia)

THE PHARYNX:

- common passage for food + air.

RESPIRATORY SYSTEM

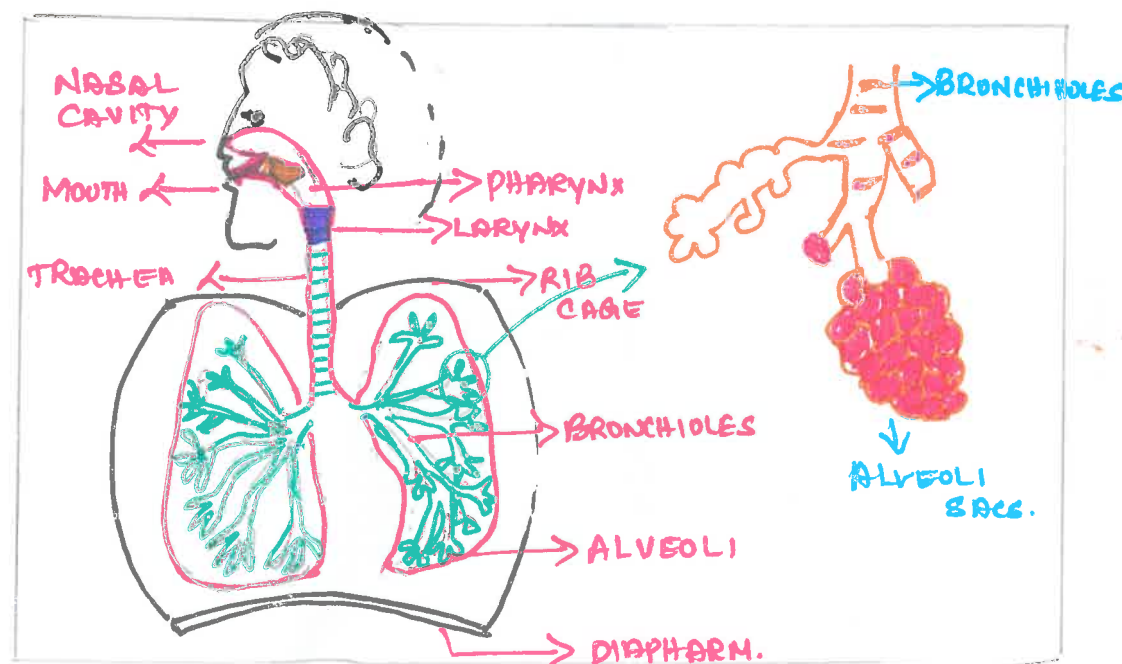


THE LARYNX:

- The larynx - voice box.
- * vibrate - make sound

THE TRACHEA:

- air → trachea - larynx.
- cartilage of rings
- inner - cilia - removes / filter dust particles.
- trachea $\left\{ \begin{array}{l} \text{left Bronchi} \\ \text{Right Bronchi} \end{array} \right.$



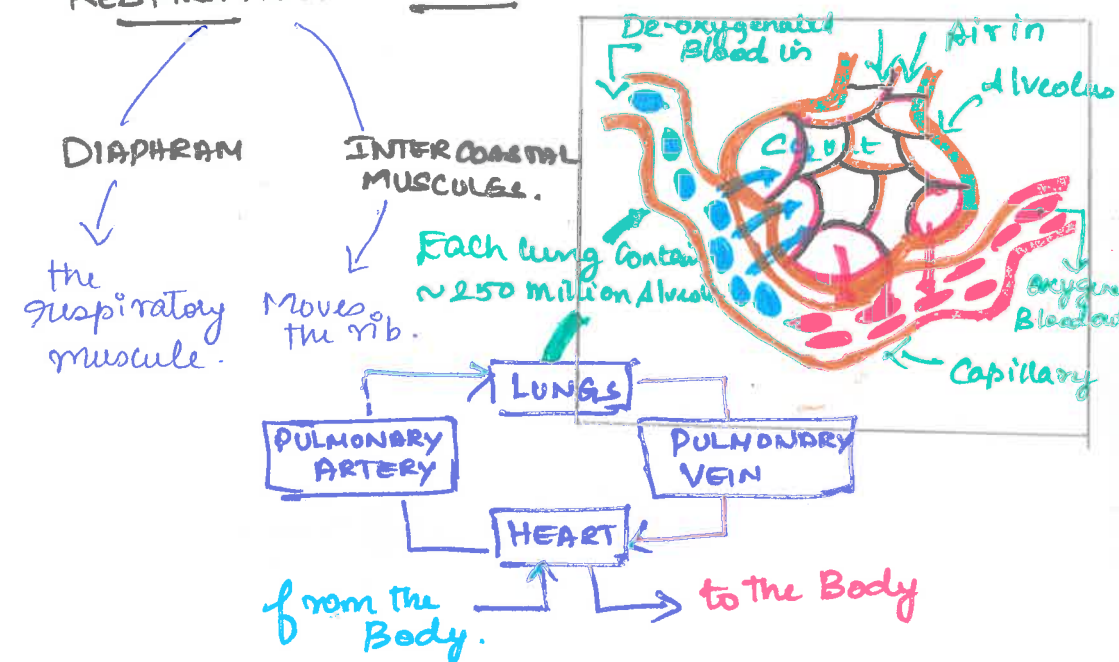
BRONCHI:

- divided into smaller Bronchioles.
- at the end - Alveoli (air sacs)

ALVEOLI:

- air sacs.
- exchange of gas takes place.
- 100 millions of alveoli
- Blood $\leftarrow O_2 \leftarrow$ alveolar air
- ↓
- rest of the Body.
- Alveoli $\rightarrow CO_2 \rightarrow$ exhaled.

RESPIRATION MECHANISM:



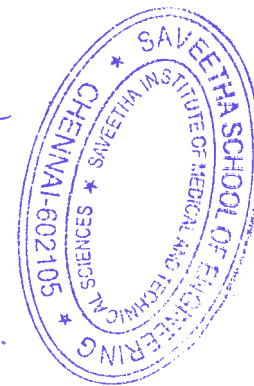
DURING INHALATION:

- * internal muscles between the ribs contract + moves up.
- * Chest cavity $\rightarrow$ increases
- * diaphragm contracts
- * Air $\rightarrow$ Nose
- ↓
- lungs. $\leftarrow$ Trachea

The inspiration and the expiration ratio varies based on:
 * the state of individual b/w rest + exhaustion
 * The depth of the respiration

DURING EXHALATION

- The rib muscles - relax
- The diaphragm - relaxes
- The chest cavity - reduced
- int. pressure $\rightarrow$ increases
- Air $\rightarrow$ lungs.



FUNCTION OF URINARY SYSTEM:

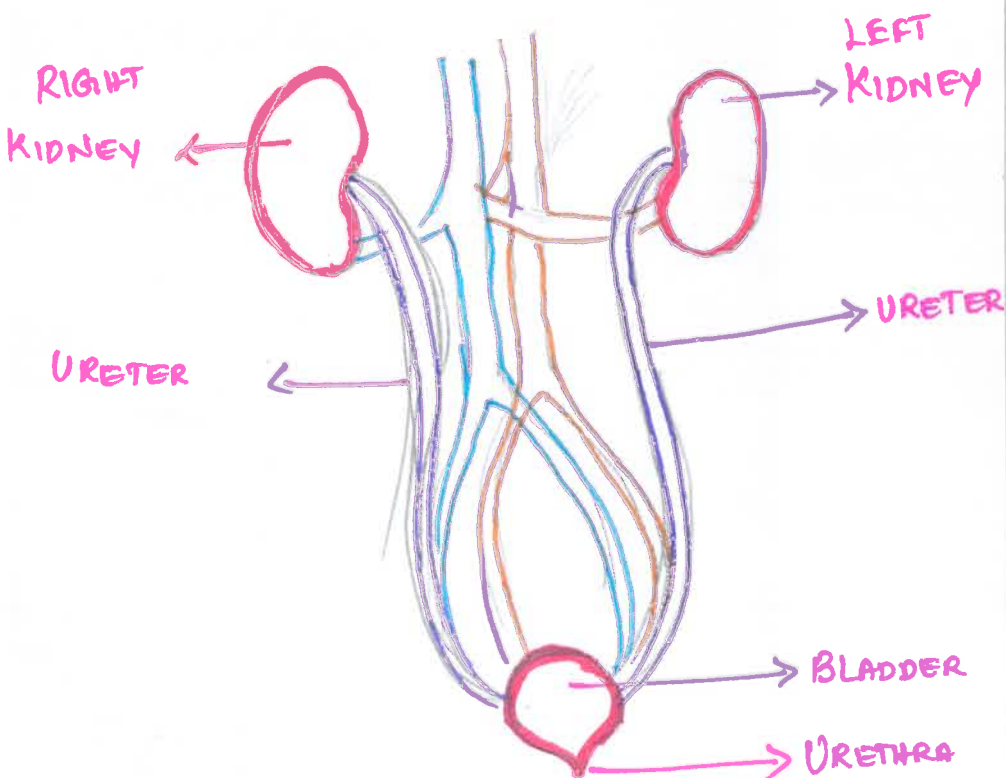
- * To filter blood and create urine as a waste by-product.

ORGANS OF URINARY SYSTEM:

- * Kidneys
- * renal pelvis
- * Ureters
- * Bladder
- * Urethra

* The body takes **nutrients** from food and converts them to **Energy**.

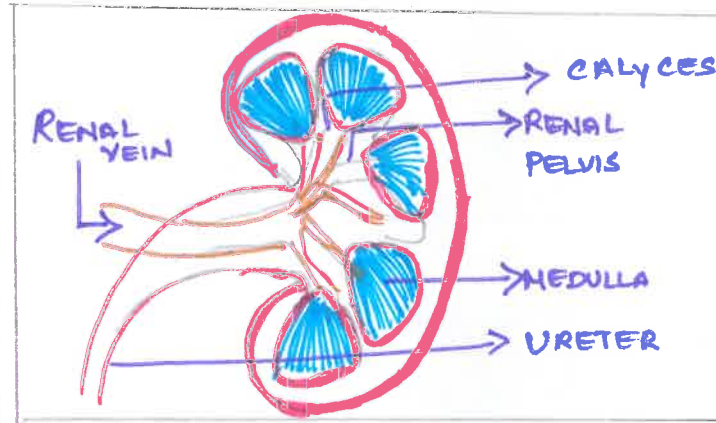
* The waste products are left behind bowel and in blood.



RENAL SYSTEM

The kidney + urinary s/m help the body to eliminate the liquid waste called **UREA**. **UREA** is carried in the blood stream to the kidneys, where it is removed along with water and other wastes in the form of urine.

ANATOMY OF THE KIDNEY:



The kidney, ureters, bladder, urethra are → urinary s/m.
→ filter, store + remove liquid waste.

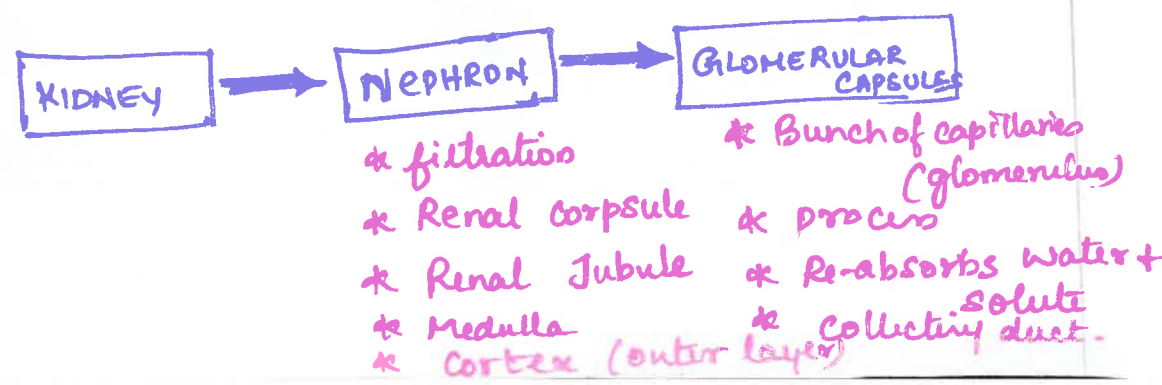
KIDNEYS: * filter blood, * eliminates the waste * make urine. * Back of abdomen.

URETERS: * two thin tubes * carries urine from your kidney to your bladder.

BLADDER: → hollow (muscle) * shape → balloon * holds → 2 cups * bladder expands.

URETHRA: * carries urine → bladder ends in an opening of your body.

MECHANISM OF FILTRATION:



URINE PRODUCTION:

1) FILTRATION:

- Blood forced through a membrane.
- only tiny particles pass through

2) REABSORPTION:

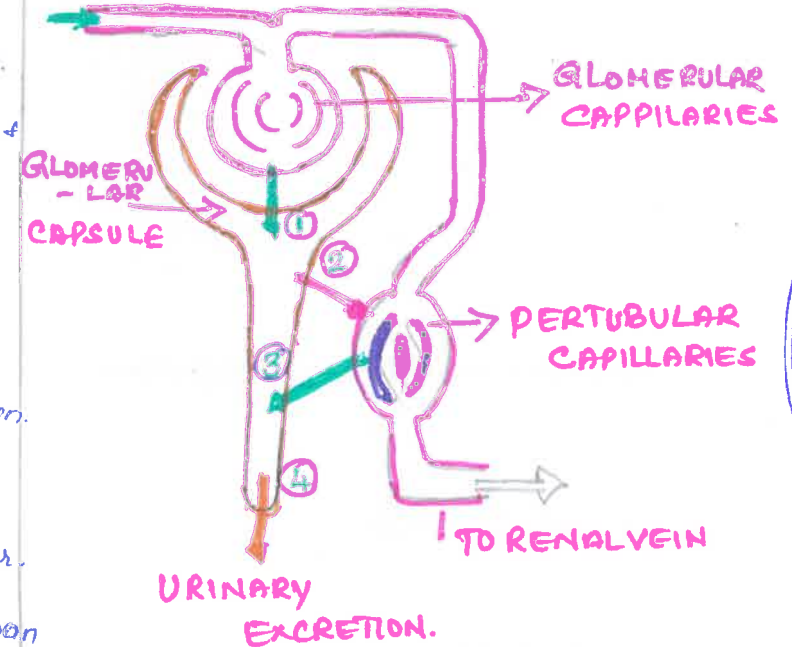
- Water + glucose go back to Blood.
- all the waste will become urine.

3) SECRETION:

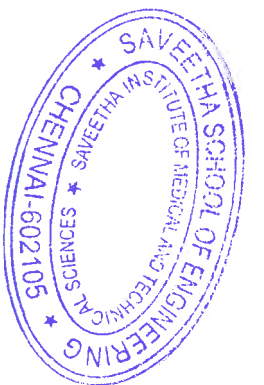
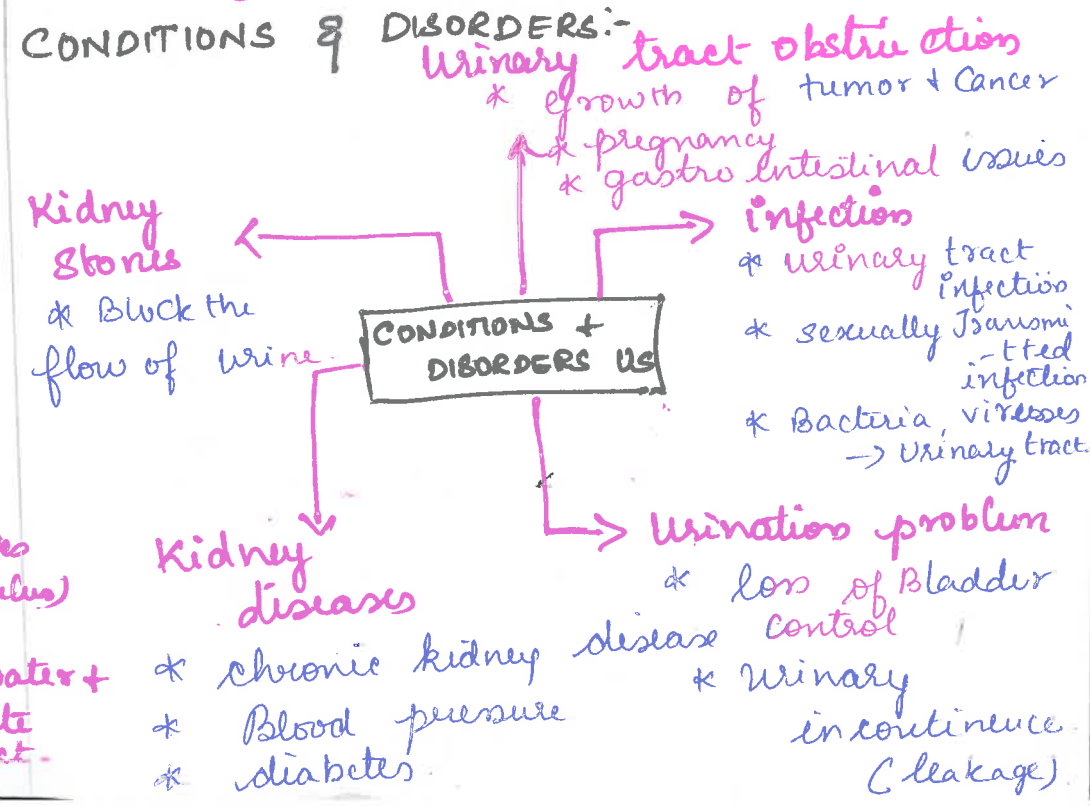
- components of blood join filtrate.
- drugs / metabolites / potassium etc.

4) EXCRETION:

- Water + urea / nitrogenous waste.
- also includes ionic solutes.

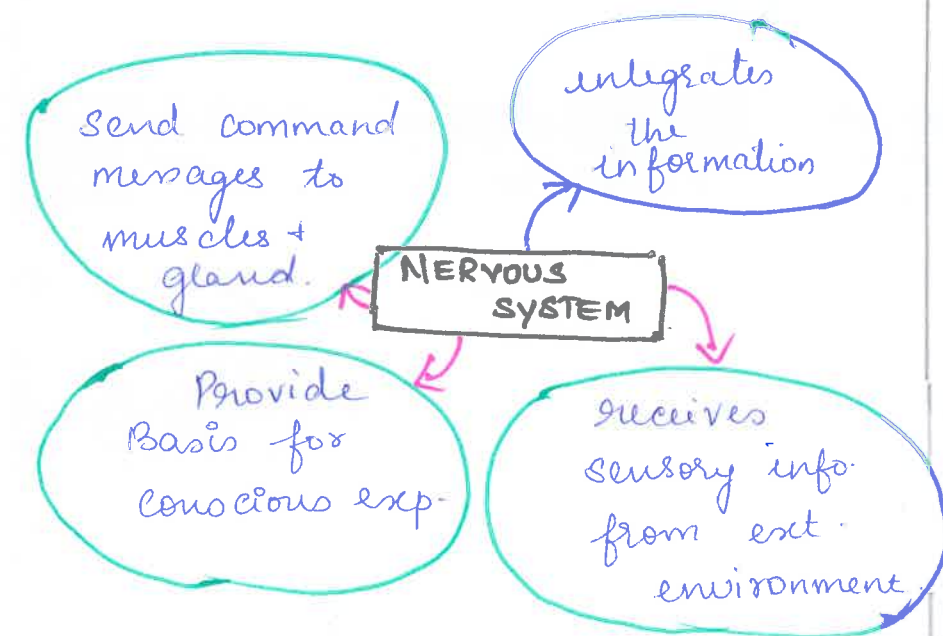


CONDITIONS & DISORDERS:-



1. INTRODUCTION:

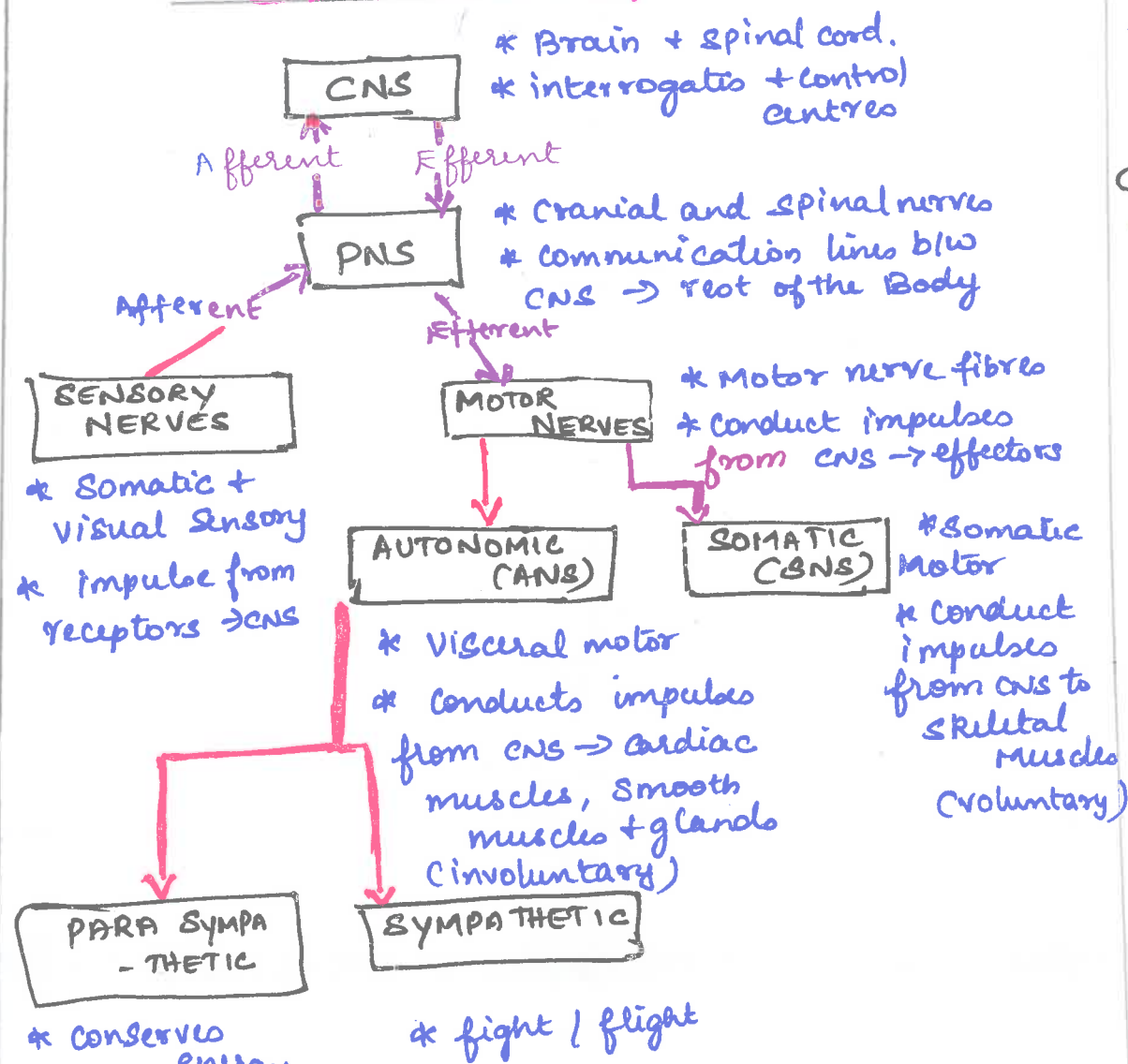
- Controls various complex functions
- co-ordinates
- composed of - Brain, sensing devices + high speed communication n/w.
- responsible - learning, memory, acquiring, interaction with society + environment.



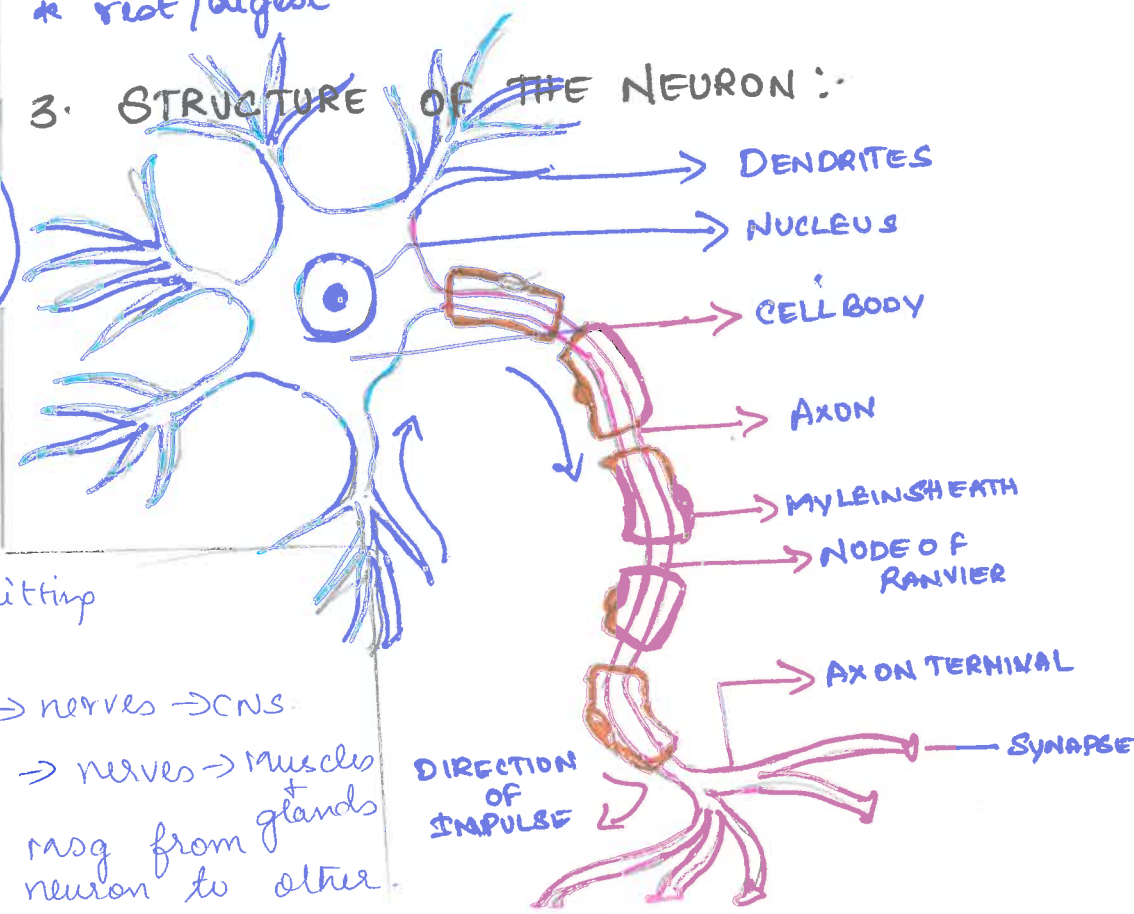
2. ANATOMY OF THE NERVOUS S/M:

- * Neuron - Basic unit.
- * Single cell - cell body - soma
- * Dendrites, Axon - long transmitting fibre.
- * Sensory - neuron → info → SO → nerves → CNS
- * Motor - neuron → info → CNS → nerves → muscles + glands
- * Inter - neuron → CNS → carry msg from one set of neuron to other

Nervous System



3. STRUCTURE OF THE NEURON:



DENDRITES:

- receive message
- allow transmission of messages to the cell body.

CELL BODY:

- Each neuron has
- * cell body with nucleus
- * Golgi Body
- * Endoplasmic reticulum
- * mitochondria

AXON

- * tube-like structure
- * carries electrical impulse
- ↳ Cell Body - Axon - other neuron
- * Surrounded by myelin sheath.

SYNAPSE:

- * chemical impulse
- one neuron terminal → dendrite of other neuron.

HOW DO NEURONS WORK?

- Sending chemicals - neurotransmitters
- from one neuron → dendrite of other neuron
- exchange of information - synapse.
- Neurons → CNS + PNS to report info.
- information enables us to think, talk, feel, move + do everything that our bodies are capable of doing.

NEURON FUNCTIONS:

CHEMICAL SYNAPSE:

- Action potential → Axon → postsynaptic ending
- releases chemical messengers. → neurotransmitters.

ELECTRICAL SYNAPSE:

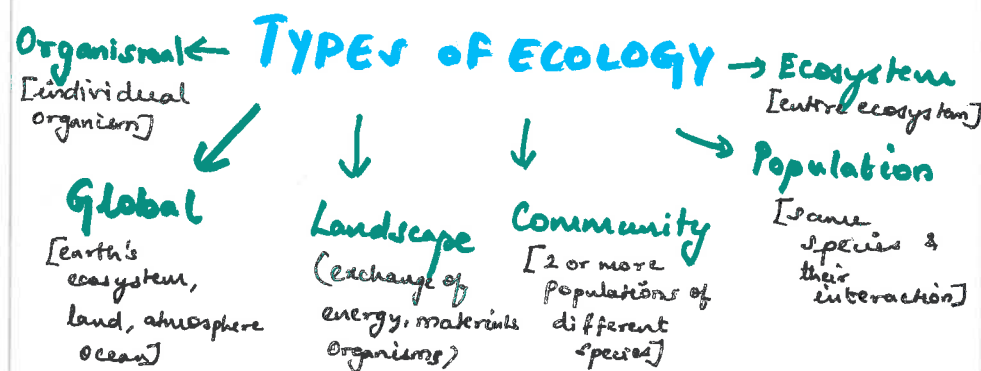
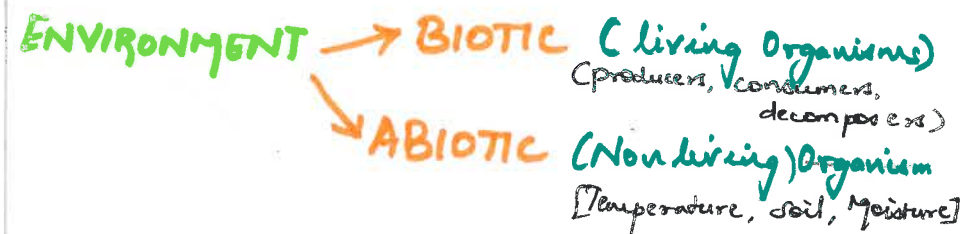
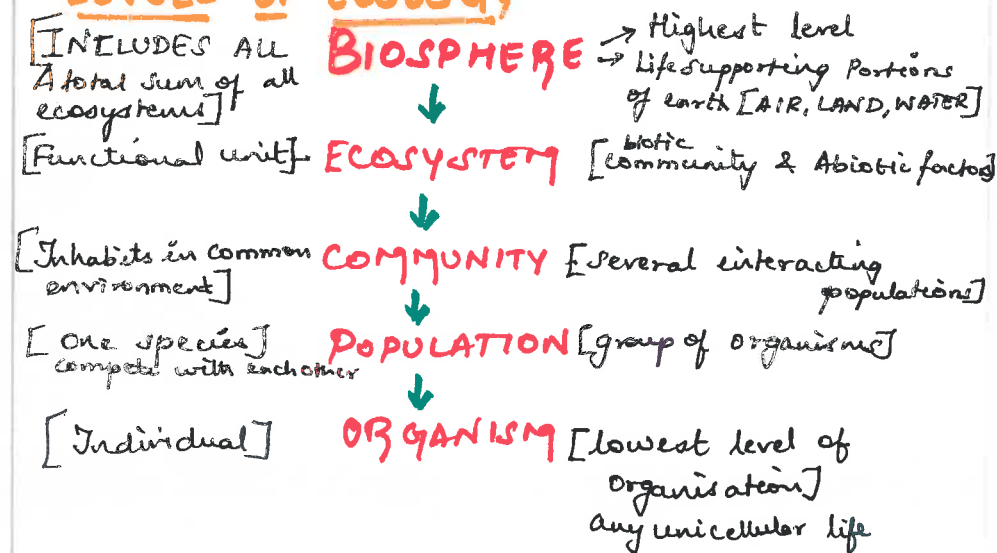
- neurons are connected by a gap junction - result in an electrical synapse
- faster than chemical synapse.



FUNDAMENTAL CONCEPTS & PRINCIPLES OF ECOLOGY

- Ecology is a branch of science, including human science, population, community, ecosystem and biosphere.
- Ecology is the study of organisms, the environment and how the organisms interact with each other and their environment.

LEVELS OF ECOLOGY



HABITAT

→ place in which the organisms lives (address)

Vs

NICHE

→ the role species plays in a community (Job)

PRINCIPLES OF ECOLOGY:

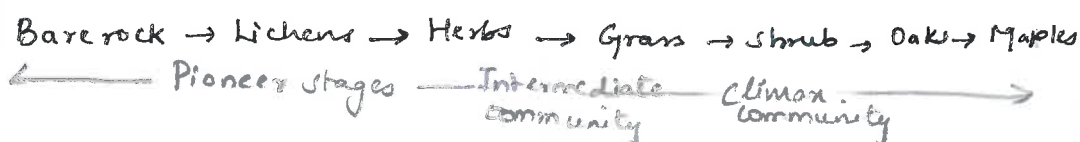
- All living organisms and their environment are mutually reactive affecting with each other
- Environment is a complex of several inter-related factors and is dynamic i.e. varies with space and time.
- Species make effort to maintain uniformity [structure, function, reproduction, growth & development]
- Organisms too modify their environment
- Both Biotic and abiotic components make up the whole biosphere [ecosystem]
- Sustained life on earth is a characteristics of ecosystem.

ECOLOGICAL SUCCESSION

- A gradual process of change and replacement of the types of species in a community

Types → Primary (process of creating life in an area where no life existed earlier) Pioneer species (Bark, Flood)

Secondary (process of reestablishment that follows a disturbance in an area, where life has formed an ecosystem).



ENVIRONMENT refers to all the conditions that influence and affect the development and sustainability of life of all organisms present on earth.



Importance of Environment:

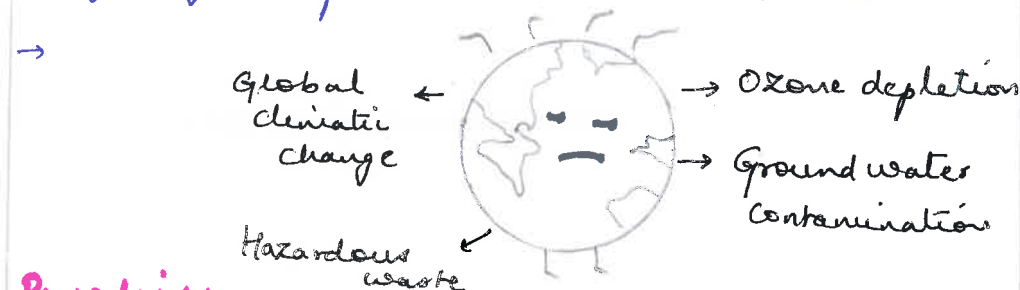
- **ENVIRONMENT** → Important role in healthy living and existence of life on earth
- Home for all living species dependent on environment for food, air, water & others.



Hence, it's so important to **Save & protect** our environment.

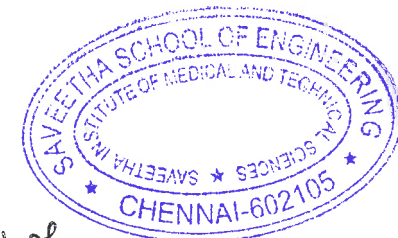
Need for Public Awareness

- Our environment is presently degrading due to pollution, deforestation, overgrazing, rapid industrialisation, etc.



Remedies:

Reduce
Reuse
Recycle



Conservation
Conserving available resources.

Control

(*) Harmful substances (BAN)

Production
Attempting Innovation

SAVE ENVIRONMENT

Preservation

Preserving the leftover

Remedies to prevent losses

Prevention

Factor harmful to the environment.

Awareness

Education
Guidance
Organisation
Participation

Biodiversity Conservation

Definition

"Biodiversity Conservation refers to the protection, upliftment and management of biodiversity in order to derive sustainable benefits for present and future generations."

Biodiversity Conservation has

Three main objectives:

- (i) To preserve the diversity of species
- (ii) Sustainable utilization of species and ecosystem
- (iii) To maintain life-supporting systems and essential ecological processes.

In-situ Conservation

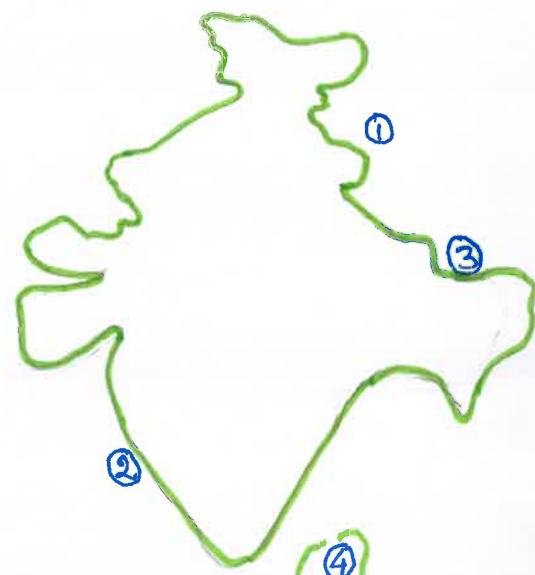
1. National Parks
2. Wildlife Sanctuaries
3. Biosphere Reserves

Biodiversity Hotspots in India

Out of the 36 biodiversity hotspots in the world, 4 are in India

These hotspots are

- (i) Himalayas
- (ii) Western Ghats
- (iii) Indo-Burma region
- (iv) Sundaland



Biodiversity Conservation Methods

Ex-situ Conservation (Artificial ecosystems)

1. Zoos
2. Nurseries
3. Botanical gardens
4. Gene banks

Biodiversity refers to the variety of living species on Earth, including plants, animals, bacteria and fungi."

The term biodiversity hotspot was first coined by Norman Myers in 1988.

Recognised tropical forests as "hotspots" according to the level of plant endemism and high level of habitat loss.

It must contain at least 1,500 species of vascular plants ($>0.5\%$ of the world's total) as endemics

It has to have lost $\geq 70\%$ of its original habitat.

1 Himalayas

Himalayan hosts almost 163 endangered species including one-horned rhinoceros, wild Asian water buffalo and as many as 45 mammals, 50 birds, 12 amphibians, 17 reptiles, 3 invertebrates and 36 plant species.

2 The western ghats

These hills are found along the western edge of Peninsular India.

Around 77% of the amphibians and 62% of the reptiles are endemic

3 Indo-Burma Region

This region consists of various countries including North-Eastern India, Myanmar and China, Vietnam, Cambodia and Thailand.

Almost 13,500 plant species can be spotted in the region, half of which are endemic and cannot be found in any other place in the world.

4 Sundaland

This region lies in South-East Asia.

Nicobar region represents India in this hotspots.

UNESCO declared this region as the world biosphere reserve in 2013.

Endangered and endemic Species

Endangered Species

(i) These are species that are on the verge of becoming extinct.

(ii) If the unfavorable conditions that have led to a population drop continue to function, such species' survival will be problematic.

(iii) Habitat loss, invasive species, and poaching are all potential threats to endangered species.

Ex: The blue whale, Indian rhino

Endemic Species

Endemic species is that ecological state of a species where a species is unique to a defined geographical location

Ex: Asiatic lion in Gir forest
purple frog in Western Ghats



ROLE OF INFORMATION TECHNOLOGY IN ENVIRONMENT AND HUMAN HEALTH

Information Technology (IT)

- * Application of Computers and telecommunication and their Equipments
- * Used to store, retrieve, transmit and manipulate data
- * Spans industries
 - Computer hardware
 - Software electronics
 - Semiconductors
 - Internet
 - Telecom equipment
 - e-commerce
 - Computer Service

IT in Environment

Tools:

- Database
- The National Management Information System (NMIS)
- The Environmental Information System (ENVIS)
- Remote Sensing
- Geographical Information System (GIS)
- World Wide Web (WWW)

Databases:

- * Collection of Inter-Related data on Various Subjects
- * Can be retrieved when required
 - Wildlife database
 - Forest cover database
 - Conservation database

NMIS:

- * Developed by DST
- * Compiled and Maintains a database on Research and development Projects
- * Keeps the record up to date

ENVIS

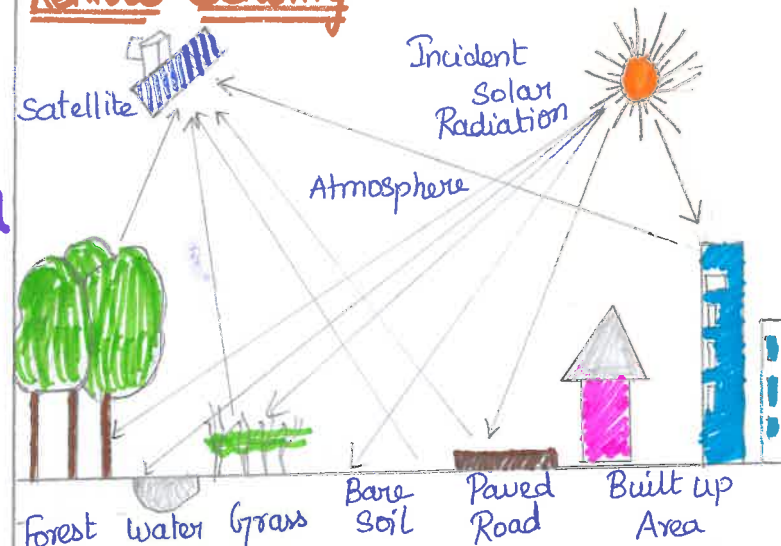
- * Established in 1982, by Ministry of Environment and Forest Government of India
- * Provides environmental Information

- policy makers
- engineers
- Scientist

To examine

- Pollution control
- Clean Technology
- Biodiversity
- Renewable resources
- wild life
- environmental Management
- Remote Sensing Technology

Remote Sensing:



GIS:

- * Used to capture, store, manipulate, analyse, manage, and present all types of geographical data
 - forest land
 - water Resource
 - Soil type
 - crop land



WWW:

- * Resources available
 - Documents
 - Audio
 - Video format
- * URL → Uniform Resource locators
- * HTTP → Hypertext Transfer Protocol
 - Web browser
 - Web Server

HUMAN HEALTH (FREE FROM ILLNESS)

Tools:

- Databases → Storing data → birth rate → Death rate → DNA database → Medical records - Malaria, Asthuma, Cancer, AIDS → Biometrics

- * NCBI - National Centre for Bio technology Information.
- * PDB - Protein Information Bank

Biometrics:

- * Unique finger Impression
- * Type of access Control and Identification
- * Used to recognise People.

Telemedicine:

- * helps to overcome the distance and saves individuals in critical and emergency condition

Broadcasting:

- * Increases knowledge about diseases
- * Specifies Preventive measures
- * Controlling infectious diseases
- * Media Spread public awareness

Conclusion:

- * With the help of IT, we can predict the Ongoing changes/predict the near future.
- * The efficiency of Patient Safety is increased by improving the living Standards.
- * Rapid growth in IT have a Positive Impact on the earth's environment and human health



ENVIRONMENTAL POLLUTION

→ Addition of contaminants to environment
↓ Effecting

Nature, natural resources and mankind

Pollutant: Substance causing harmful effects on organisms

CLASSIFICATION OF POLLUTANTS

Persistent Pollutants: Remain in environment → long time → original form
Non Persistent Pollutants → Break down to simple form by living organisms → biodegradable pollutants

Another classification

- Primary Pollutant: Remain in the form in which they were added.
- Secondary Pollutant: Formed due to interaction with primary pollutants

Based on origin

- Man made Pollutants
- Natural Pollutants

Based on nature of disposal

- Biodegradable
- Non biodegradable

TYPES OF ENVIRONMENTAL POLLUTION

→ Air Pollution, water pollution, soil pollution, noise pollution

POLLUTION - CAUSES, EFFECTS AND CONTROL MEASURES

AIR POLLUTION

(i) Causes of Air Pollution

- ↳ Burning of Fossil Fuels
- ↳ Emission of harmful gases from automobiles
- ↳ Emission of gas from factories and industries
- ↳ Mining Activities
- ↳ domestic resources like ACs, chemical paints etc.
- ↳ Agricultural: insecticides, fertilizers, pesticides

(ii) Effects of Air Pollution

- ↳ Global warming
- ↳ Acid Rain
- ↳ Ozone layer depletion
- ↳ Effect on animals and aquatic life
- ↳ Problems in lungs, respiratory systems
- ↳ Irritation in lungs, throat, nose, allergies
- ↳ Kidney problems, hypertension

(iii) Control measures

- ↳ Minimize using vehicles - Public transport
- ↳ Use of clean energy resources
- ↳ modification of industrial processes
- ↳ conversion of pollutants → less toxic form

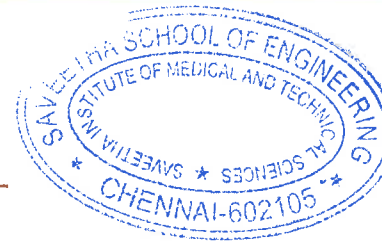
WATER POLLUTION

(i) Causes

- ↳ Industrial waste water → Fertilizers, Pesticides etc.
- ↳ Agricultural sources, thermal pollution → thermal power plants
- ↳ Marine pollution → oil spills, river discharges
- ↳ Sewage waste into water bodies
- ↳ Mining waste

(ii) Effects

- ↳ Excessive mercury → Minamata disease
- ↳ Lead → dyslexia, Cadmium → Itai-Itai disease
- ↳ Polluted water → low DO → eliminates sensitive organisms
- ↳ Biological magnification and eutrophication
- ↳ Excess nitrate → dangerous for human health
- ↳ excess fluoride → Teeth deformity, neuromuscular disorders



(iii) Control measures

- ↳ Recycling and treatment of water
- ↳ Minimized wastewater discharge
- ↳ Organic farming, minimized fertilizer use
- ↳ Reduce water consumption

SOIL POLLUTION

(i) Causes:

- ↳ Industrial waste
- ↳ Agricultural waste: Fertilizers, pesticides etc.
- ↳ Discarded materials: Radioactive, Plastic etc. elements, bags

(ii) Effects of Soil Pollution

- ↳ Reduced soil fertility
- ↳ Ecological imbalance - Imbalance in flora & fauna
- ↳ Increased soil erosions
- ↳ Clogging in drains, release of gases, problems in waste water management

(iii) Control Measures

- ↳ Afforestation
- ↳ Solid waste management
- ↳ Minimize plastic bag usage → biodegradable products
- ↳ Biomedical waste → Incinerated

NOISE POLLUTION

Sources

- Indoor → TV, generators, etc.
- Outdoor → construction, social events, loudspeakers, industrial activities, automobiles, rail traffic, aeroplanes

30 dB → indoor
75 dB → hearing damages

Effects: Sleep disturbance, hypertension, hearing difficulties, cardiovascular problems, Sleep disorders, aggressive behaviour, stress etc.

Control Measures:

- ↳ Large sounds in public places like sirens → Banned / Controlled.
- ↳ Use of explosives → minimized
- ↳ Sound proofing equipments like generators

URBAN PROBLEMS RELATED TO ENERGY - RENEWABLE ENERGY - WASTE (WATER) MANAGEMENT & PERSONAL HYGIENE

Energy

- Many Primary energy sources - nuclear, fossil (oil/coal/natural gas), and renewable (wind/solar/geothermal/hydropower) are → electricity (secondary energy source) power lines & other transmissions for home & business.

Urbanization & Associated Energy Problems

"Urbanization":

population (Rural areas) → urban areas (towards towns & cities)
→ in search of employment
✓ better standard of living

Resulting in expansion of cities
"URBAN SPRAWL"

"Energy Requirements":

- ✓ 75% of produced energy → urban areas
- ✓ for transportation, buildings, industries, plants (municipal) & agriculture.

"Deficit & Demand": (India)

- ✓ Production: 1,383 TWh (fossil: 75.38%, Renewable: 21.26%)
- ✓ GHG emissions: 2,308 Mt of CO₂
- ✓ Average electricity/capita: 1,208 KWh
- ✓ Distribution losses: 20.66%

"Alternative Energy" - "Renewable Energy"

- ✓ Energy from → non-depleting source
- ✓ sunlight, wind, tides, geothermal, hydro, biomass
- Solar Energy**: Light → electricity (☀ photovoltaics)
- Wind Energy**: wind → electricity (☪ wind → turbine → generator)
- Hydro Energy**: moving water → electricity (☪ water flow → turbine → generator)
- Geo-Thermal Energy**: heat from earth → electricity (☪ heat → steam → turbine → generator)
- Bio-mass Energy**: organic matter → electricity (☪ organics → burnt/combusted → generator)

WASTE - WASTE WATER - MANAGEMENT & PERSONAL HYGIENE

Waste - Waste water - Management

- mainly generated from industries, domestic & agricultural activities
- needs to be → identified → managed

Types of wastes

- Solid** - any garbage, sludge, plastics, wood, food etc,
- Hazardous** - any ignitable, corrosive, explosive solid / semi solid / liquid
- Universal** - ✓ agricultural run-offs ✓ batteries
- e-waste** - ✓ computer & spares / components

Effects

- difficult recycling - non-biodegradable
- dumping → infertile soil → putrefaction
- health hazards from exposure
- human/waste density ratio will be increasing

Management

- Separate & segregate
- Reduction, Recovery, Reuse, Recycle.

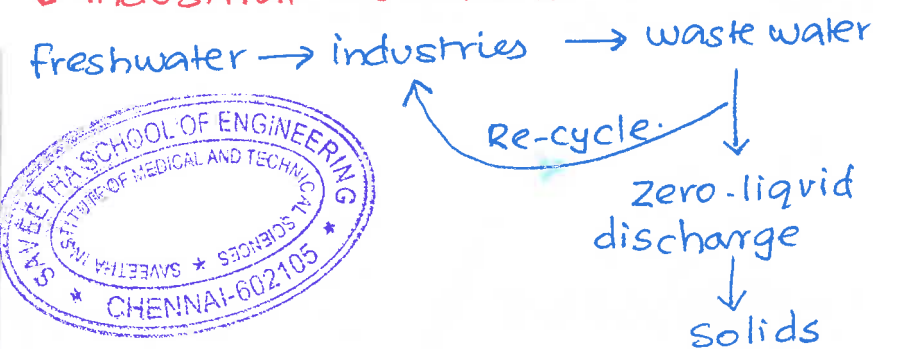
Wastewater

- water generated after the use of fresh water, raw water, drinking water or saline water
- used water from any / combination of domestic, industrial, commercial, agricultural sources
- run-off / storm water, sewer inflow

Sewage treatment

raw sewage → primary effluent.
↓ grit disposal
primary effluent → Aeration tank → Activated sludge
↓ discharge to surface
(or) Tertiary treatment → disinfectant

Industrial treatment



Leachate treatment

wastewater → matter → treated water
↓ extracted
Soluble / suspended solids
Eg:- Landfill leachate treatment.

Management

- ✓ Primary - Physical screening / sedimentation aeration
- ✓ Secondary - aerobic, anaerobic, composting (biological)
- ✓ Tertiary - Chemical treatment, eg. chlorination

Personal Hygiene

- ✓ cleaning your body every day
- ✓ washing your hands with soap
- ✓ brushing your teeth twice a day
- ✓ covering while sneezing / coughing

Why needed?

- ✓ prevents spreading germs & infectious diseases
- ✓ maintains health
- ✓ avoids food contamination
- ✓ prohibits from becoming a vector

WASTE & WASTEWATER MANAGEMENT

Waste Management

- ✓ Solid waste - all non-liquid wastes
- ✓ Associated risks if not properly managed.

- Disease transmission
- Pollution
- Unhygienic & untidy environment

✓ Categories of solid wastes

- Organic wastes - degradable wastes especially food
- Combustibles - paper, wood, dried leaves etc,
- Non-combustibles - metal tin cans, bottles, stones etc,
- Ashes/dust - residues from fire, industries soot
- Bulky wastes - tyres & rubber wastes
- Hazardous wastes - oil, battery acid, medical wastes
- Construction wastes - roofing, rubble, broken concretes

✓ Effects if not managed properly

- Difficult in recycling
- Dumping in excess leads to infertile soil or putrefaction
- Health hazard from exposure
- Human/waste density ratio will be decreasing

✓ Management

Identify type of waste

Identify source of waste

Determine potential health hazards from waste

Determine the volume of waste generated

Identify safe collection methods

Identify safe transportation methods

Identify safe disposal methods

Separate & segregate

✓ Key components of solid waste management

- Generation
- Storage
- Collection
- Transportation
- Disposal

• **Generation** - materials - valueless to owner and thrown off

• Storage

- ✓ small containers - bins, household containers
- ✓ large containers - drums, communal bins
- ✓ shallow pits
- ✓ communal depots - fenced/walled areas

• Collection

- ✓ Collection intervals & volumes of collected wastes must be estimated carefully.
- ✓ cycles, trucks etc., are used for transport

• Transportation

- ✓ Human-powered: hand carts, tricycle etc.,
- ✓ Animal-powered: donkey carts, etc.,
- ✓ motorised: tractors, trucks etc.,

• Disposal

- ✓ Burial or land-filling
- ✓ Composting
- ✓ Burning or incineration
- ✓ Recycling

Wastewater Management

• water generated after the use of freshwater, raw water, drinking water or saline water.

• used water from any/combination of domestic, industrial, commercial, agricultural sources

• run-off / storm water, sewer inflow

✓ Sewage treatment - wastewater treatment plant.

→ sewage is complex of mixture

- suspended solids
- organic impurities
- inorganic impurities
- nutrients
- microbes

COD
BOD
Chemical
Biological } oxygen demand

Raw sewage → Primary Effluent
WWTWP. Grit Disposal

Primary Effluent → Aeration tank
Discharge to surface ← Activated Sludge

(or)
Tertiary treatment

✓ Industrial Treatment

Fresh water → Industries → wastewater
Recycled water ↓ WWTWP
zero-liquid discharge
↓
solids.

✓ Leachate treatment

wastewater → Leachate matter → Treated water
extracted soluble/suspended solids

✓ Management - Treatment categories

- Primary - physical screening/sedimentation
- Secondary - aeration (aerobic/anaerobic), (biological) composting
- Tertiary - chemical treatment

Personal Hygiene

• Oral, hair, foot, hand, home } cleanliness

- ✓ cleaning your body everyday
- ✓ washing your hands with soap
- ✓ brushing your teeth twice a day
- ✓ covering while sneezing/coughing
- ✓ maintaining sanitation

• Why needed?

- ✓ prevents spreading germs & infections
- ✓ maintains health
- ✓ avoids food contamination
- ✓ prohibits from becoming a vector

Reduce

- * In general reduce refers to minimize the amount of waste that we create.
- * For eg: let us consider the primary source of our living is nothing but water.
- * As per survey only 3% of water is good.
- * Hence it should not be wasted.
- * Plastic affects the environment globally & 56 lakh tones of plastic is accumulated every year.
- * These plastics end up in river, mountain, lands etc.
- * In recent studies it is revealed that even fishes have small microns of plastic inside them & when we consume it our health is also affected.
- * Actually most of plastics are not recycled & hence it must be avoided or atleast reduced.

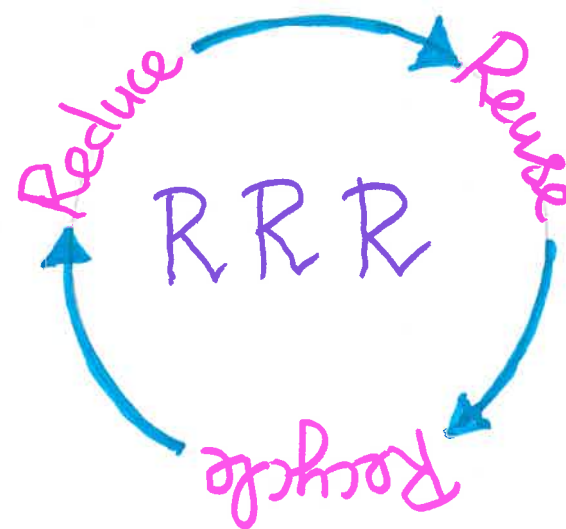
Electricity Instead of using conventional electricity, we can go for solar which is a renewable source of energy.

Also we can reduce the consumption of electricity by switching off electrical appliances i.e. fan, light, ac, laptop, desktop etc.

Reuse

Reuse $\Rightarrow$ using things again.

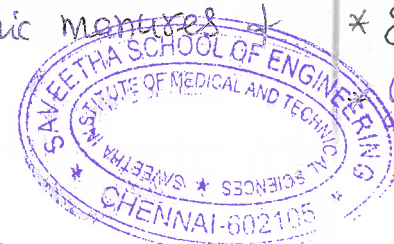
- * In order to manage plastic waste we can reuse plastics.
- * Instead of throwing away plastic bags, bottles, buckets etc it can be reused again in a lot of creative ways.
- * We can make **DIY** (Do it yourself) using different plastics.
- * For eg water bottles can be used for garden planting etc.
- * These reusing of plastics will help our environment.
- * Sewage water can also be recycled and used for plants.
- * It also reduce green house gas emission and helps in sustaining the environment.



Recycle

Recycle $\Rightarrow$ use same materials to make new products.

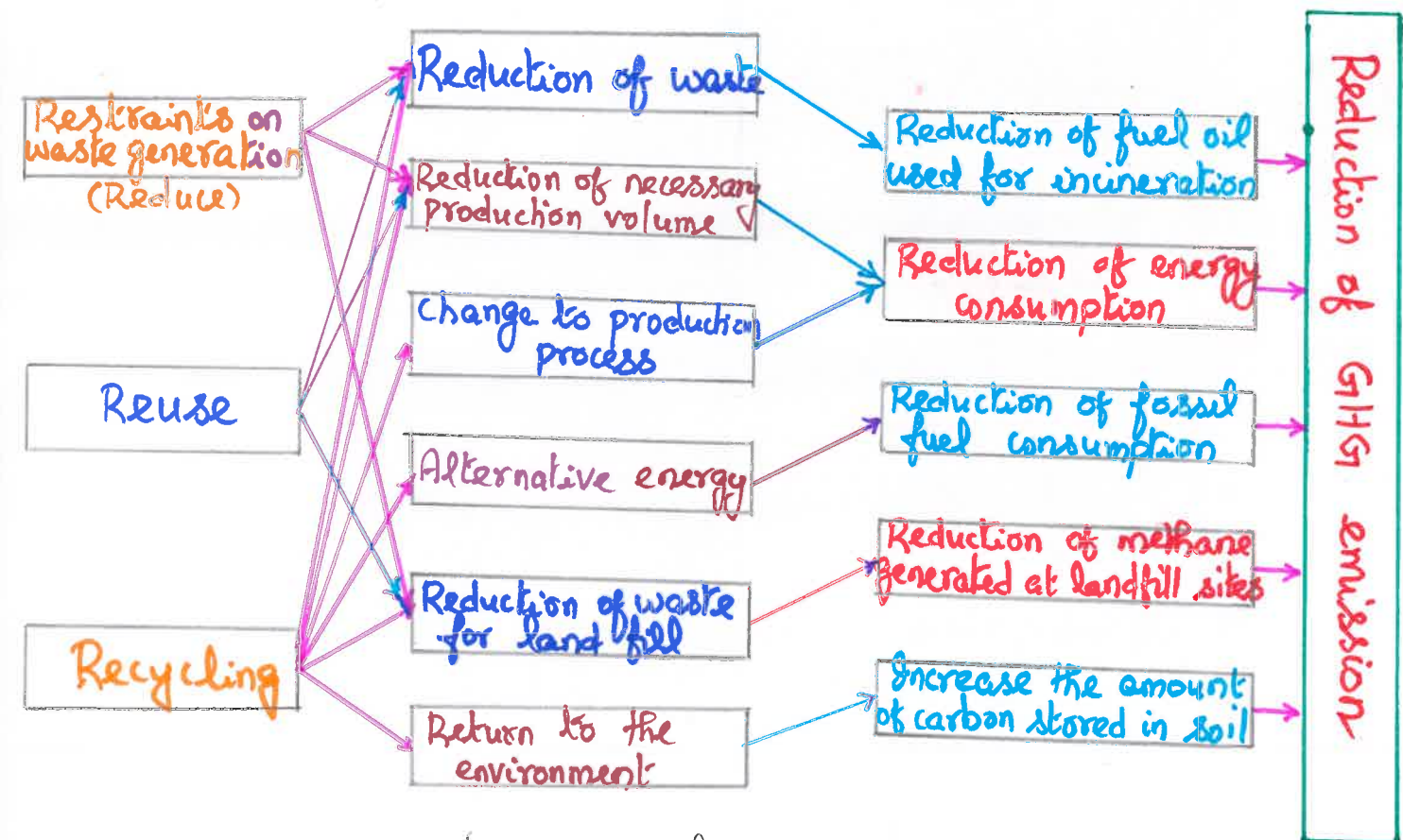
- * It is a preventive measure for saving useful materials from wastage & also protecting the environment.
- * The biodegradable waste like food particles, kitchen wastes etc. can be turned into organic manures & fertilizer.



7 Elements to improve Eco efficiency.

- * Reduce material intensity
- * Reduce energy intensity
- * Reduce dispersion of toxic substances
- * Enhance the ability to recycle.
- * Maximize use of renewable resources
- * Extend product durability.
- * Increase service intensity.

3R Reduces Green House Gas Emission



- * To sum up it is concluded that the **3R** (Reduce, Reuse & Recycle) is one of the best ways for preserving the natural resources, reducing pollution & help us to make our environment a better place to live in for us & our future generation.

WASTELAND RECLAMATION

Wastelands

- Economically unproductive lands

Suffering from environmental deterioration.

Categories of Wastelands

- * Gullies / ravines
- * Upland with / without scrub
- * Waterlogged & Marshy land
- * Land affected by salinity / alkalinity
- * Land under shifting cultivation
- * Degraded forest / grazing land
- * Mining / industrial wastelands
- * Barren rocky / stony waste / sheet rock areas
- * Snow covered / glacial area.

More than $\frac{1}{2}$ of our country's geographical area

- About 175 Million ha.

Maximum (36 Million ha) @ Rajasthan

Causes of Land degradation

- * Over cultivation
- * Deforestation
 - Use of fuel wood
 - Shifting to cultivation
 - Commercial timber exploitation
 - Clearing for human settlement, industries
- * Over grazing
- * Improper irrigation practices.

Effects of Land Degradation

- * Surface runoff & floods
- * Soil erosion & desertification
- * Loss of nutrients & land productivity
- * Soil acidification / alkalisation
- * Soil salinity
- * Loss of biodiversity
- * Long term socioeconomic impact on humans

Wasteland reclamation is the process of turning barren, sterile wastelands into fertile & suitable for habitation & cultivation.

Objectives of WLR:

- * To improve the physical structure & quality of marginal soils.
- * To improve the availability of good quality water for irrigating these lands
- * To prevent soil erosion, flooding & landslides
- * To conserve biological resources of the land.

Wastelands can be classified into 3 forms:

- i) Easily reclaimable
 - Reduce salt content & used for agriculture
- ii) Reclaimed with some difficulty
 - Used for agro forestry
- iii) Reclaimed with extreme difficulty
 - forestry or to recreate ecosystem



Important Reclamation Practices:

1. Afforestation :-

- Growing forests over culturable wet wasteland
- National Commission on Agriculture (NCA) & the National Wasteland Development Board (NWDB) set a target of 5 million ha wasteland to forest.

2. Reforestation :-

- Replanting over lands destroyed by fires, over-grazing & excessive cutting.

3. Providing surface cover :-

- Leave crop residue on the land after harvesting.

4. Mulching :-

- Protective cover of organic matter & plants like stalks, cotton & tobacco stalks, etc.

5. Changing ground topography on Downhill's :-

- a) Strip farming - Different crops in strips
- b) Terracing - Earth is shaped into levelled terraces
- c) Contour ploughing - ploughing done across hill

6. Leaching :

- Salinity reduced by filling 0.5 to 1.0 cm of water in banded small plots.

7. Changing agricultural practices:

- Mixed cropping, crop rotation

8. Ecological succession :

- Natural development / redevelopment of an ecosystem.